Master Degree Project



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Achievement motivation in video games

A Study of Player Motivation Through a Case Study and Qualitative Interviews

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Abstract

Nowadays, achievements are a common feature in video games because they can motivate players. However, less attention has been given to understanding how achievements influence player motivation, therefore, this paper's goal is to explore what motivates players to continue playing a game in order to collect achievements. This study reviews existing literature on achievements, including academic research and a case study of the popular gaming platform Steam, to identify key motivational factors. These findings provide the foundation for qualitative interviews with players who are highly motivated to complete achievements, exploring the motivations and emotions associated with achievement hunting. The results identified four main themes: player motivations, player experiences, social influences, and achievement design factors.

Keywords: Achievement, Game development, Motivations

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1 Introduction

One day, I saw a friend play a video game in a particular way. They were very meticulous in how they played, always having a guide next to them for information about the game, skipping all cutscenes, and having clear goals, but not in a quick sense as in a speedrun. When I asked them about it, they explained that they were playing to get all the achievements, and that this was not their first playthrough. Their motivation lay somewhere different from what playing a game normally comes from.

An achievement in video games can be described as a virtual reward collecting system, usually separated in some sense from the rest of the game (Jakobsson, 2011, p. 1). The visual style of an achievement usually follows a similar pattern on any game platform. It consists of a name for the achievement, an icon and a description that tells a story of what the player has to do to earn it (Hamari & Eranti, 2011, p. 5-6).

The task of going for an achievement can be used to motivate a player or entice them to go for tasks in games that otherwise would be seen as boring, because it can add a layer of challenge, play, fantasy and reward to the activity at hand (Blair, 2011, p. 7-8).

Some studies made by industries have found that games with achievements generate more revenue than games that do not have achievements, as well as better critical reception. All games on the platform Microsoft's Xbox Live are also required to have achievements, leading to a clear push for games to include them, aside from Xbox 360 Indie Games and digital re-releases (Hamari & Eranti, 2011, p. 1; Jakobsson, 2011, p. 6). This also correlates with Jakobsson (2011, p. 6), with achievements having a commercial impact.

The goal of this paper is to explore what motivates players to continue playing a game in order to collect achievements. It examines the motivations behind pursuing achievements, how achievements can be designed to appeal to players, and what effects they have on players when they are obtained.

The research reviews existing literature on the subject of achievements. This latter gets followed up with a case study of the popular game platform Steam, both having a focus on achievements and motivational factors. These two forms the groundwork for qualitative interviews with participants who feel a strong urge to complete achievements, allowing for a deeper understanding of how these systems affect player experience.

A secondary goal of this research is to explore achievements from the perspective of game developers and how it might be relevant for researchers, to examine what additional purposes they may serve. Taking into consideration whether achievements can be used for purposes beyond player motivation and investigates their potential limitations. Topics include their use in representing other in-game data points, their role in serious games, and the ethical considerations surrounding their implementation. These aspects will be discussed with support from relevant theoretical frameworks.

2 Theoretical Background

2.1 An overall look on achievements

Achievement-like rewarding systems can be traced back to the 80s, where fabric patches were sent out if a player could prove they had completed feats in games by taking a picture of their screen and sending it in (Jakobsson, 2011, p. 5). If you have played any game with achievements on Steam, Xbox, or PlayStation, you might have heard a jingle melody play or seen a small popup window appear in a corner, signaling the reward when progressing in the game. The game platform might also have more information on how you received the achievement and how you can receive more of them. This is an achievement, and further down in the paper, through a case study, this paper will look more closely at how a player stumbles upon them.

Giving more context to how an achievement's visual style follows (1.) To start with the rewards *name* it is usually connected to the overall concept of the game, the game sets a theme for reward. (2.) The *icon* or the visual aspect follows a similar pattern to what the name represents, what the icon usually has is when not having received an achievement they tend to be faded out or grayed out, in the option when the achievement is even present to be viewed before receiving it. (3.) The job of the description is usually to convey to the player what is required from the player how to complete the achievement and result of requiring it. A developer can of course decide to obscure the information, and tell vague descriptions. To be able to decipher or comprehend the achievements an understanding of the game world is sometimes required to decipher the description (Hamari & Eranti, 2011, p. 5-6).

The coining of the word “achievement” and what made it popular is said to come from Xbox 360; this later spread to other platforms such as Steam and PlayStation (Jakobsson, 2011, p. 5). On Xbox 360, every game must have 1000 Gamerscores, which developers can divide into up to fifty achievements. The choice of how many points each achievement should give is up to the developer (Jakobsson, 2011, p. 3).

“According to Montola et al. (2009), achievements are optional sub-goals in a secondary reward system (achievement system). Björk (2011), in his continuing work on describing game design patterns, defines achievements as “goals whose fulfilment is stored outside the scope of individual game sessions”. Jakobsson (2011) sees achievements as quests in a system where players collect virtual rewards which are separated from the rest of the game.”

Hamari & Eranti (2011, p. 3)

According to different viewpoints that Hamari & Eranti (2011, p. 3) bring up, they follow a similar pattern of achievements being a side aspect of games, but they also present some more complicated points, for example that they are optional or secondary. This can be argued to not be fully correct when people start pursuing achievements as more than optional or secondary. They themselves also point out that achievements should be viewed as games of their own.

2.2 Player types

There is a multitude of research on the topic of player types in video games. One of the player types discussed usually concerns topics such as achiever or completionist perspectives on video games. While this research delves heavily into achievement casuals, achievement hunters, and achievement completionists (Jakobsson, 2011, p. 12–16), there are also studies indicating that one player type does

not suppress another, showing that one can fall under achievers as well as other player types, such as, for example, social and immersion components (Yee, 2006, p. 3).

Fritz and Stöckl (2023, p. 4-6) created 80 questions asking players about how much they enjoy different aspects in games and trying to find desires and motivations. They found the type groups of (1.) Social, motivated by casual socializing. (2.) Social competition is the desire to compete against other players. (3.) pushing yourself to the limit of your skill, the challenge. (4.) Escapism is about escaping reality and avoiding thinking about real-life worries, as well as change of pace. (5.) Motivated by the desire to play characters within a story, Role-playing. (6.) Power fantasy is the desire to feel powerful in a game or to play someone that is powerful. (7.) Building and creating within a gameworld, is the desiree of creation. (8.) Exploring, when the player is motivated to explore the game world for the sake of exploring. (9.) When someone wants to complete everything in a game, whether it is collectables or completing the tasks in story, this would fall under completion. (10.) Griefing, the desire to impose oneself into clothes to their detriment.

What can be interesting instead is looking at different levels of degree within the topic of achievers. (1) Casual hunters view them from a casual perspective; the achievement system adds value by adding goals to the game. The game itself comes first, and the achievements are viewed more as a secondary aspect, not until you get your first achievements are you reminded of them, or when a game is finished but you want to continue playing. The view on motivation falls somewhere where the reward matters enough that we feel motivated when receiving it, but not enough that the player feels forced to continue solely for the rewards.

(2) When achievements become more important than the game, it falls under achievement hunters, who might even find themselves playing outside their preferred genre. This could be described as grinding for achievements. In the Xbox 360 system, it could be more common to see them go for a higher accumulated Gamerscore instead of getting all achievements in a game, focusing on time-efficient ones. Hunters play less for the fun of the game, and the engaging part is instead exhibiting skill and dedication.

(3) Achievement completionists are about collecting everything that falls under the right category; this can also include items and unlockable characters. A game can be considered unfinished until they have everything, including achievements. The right game to complete is more important than it being easy to complete. Completionists can also feel a lack of fun from the activity but can find it valuable in other ways (Jakobsson, 2011, p. 12–14).

The subject around achievement types is a hard topic to find information on, but one other source more focused on school environments and game achievements was found. They studied four different types: super-achievers, mastery-only, performance-only and non-achievers. Mastery in this case is about one's own will to strive to do well and has been led to be a good way of learning. Performers in this case strive to perform better than peers, something that can lead to learning, but sometimes also hinder learning. Non-achiever and super-achiever are the cases of either no interest or interest in both performance and mastery (Heeter, et al., 2011).

Based on Heeter (2011, p. 43) where the non-achievers with 29%, were below median on both mastery and performance, mastery-only 16%, performance-only 17% and super-achievers had 38%. This data was based on 432 university students (Heeter, et al., 2011, p. 43).

What the results ended up showing was that performance ended up being more important in games, while in school mastery was more important. Some other result shown was that the people that had

goals for school also tended to have strong achievements goals for gaming, the trend being strongest for performance (Heeter, et al., 2011, p. 43).

2.3 The motivator of achievements

The brain perceives information on a regular basis. We have our perceptions of the world that we process, where attention, motivation, and emotions can influence how we process that into memory. Anything in your life is basically rewiring your brain; this includes video games and their rewards as well (Hodent, 2021, p. 6–7).

Our attention span can either be active or passive. Active, in the sense of actively controlling one's attention to direct it towards an activity, while passive is when something around oneself triggers one's attention as an additional response to active attention (Hodent, 2021, p. 13). As an example connected to the paper's goals, this could be active attention on a game and something happening in the game giving the player an achievement, with passive attention being placed on noticing the reward.

People react to different types of factors motivating them. These factors can be based on one's own interest and value, or it could be for an external reason (Ryan & Deci, 2000). When discussing motivation, there are often two sides that occur in discussion, and these are also the most relevant to the subject of achievements: extrinsic motivation and intrinsic motivation (Ryan & Deci, 2000; Hodent, 2021). These can occur at the same time and are not contradictory to each other. Extrinsic motivation occurs when you do something in order to obtain something else, in this case, for example, a rewarding achievement (Hodent, 2021, p. 15; Blair et al., 2016, p. 1). The other, intrinsic motivation, is when you do something for the pleasure of doing it, when the activity itself is the motivator. For example, in video game settings, this can look like the game itself being the intrinsic motivator, while the loot, rewards, and even punishing parts like loss and failure in the game being extrinsic motivators (Hodent, 2021, p. 15).

Ryan and Deci (2000, p. 5) divides extrinsic motivation into different categories from external factors to internal factors. (1.) External being for example rewards and punishments, (2.) somewhat external leads more to internal rewards, (3.) somewhat internal is conscious choices and personal gains, (4.) Internal is about awareness and synthesis with oneself. The next step after the extrinsic internal follows intrinsic motivation that is also internal and is about one's interest, enjoyment and inherent satisfaction. For example, the internal extrinsic motivation shares many similar qualities with intrinsics, but they are done to attain a separable outcome rather than their inherent enjoyment (Ryan & Deci, 2000, p. 6).

Operant conditioning, or instrumental learning, is about wanting to increase a behavior: you can reward the behavior, and to decrease a behavior, you can punish that behavior. What is not always taken into account are the emotions that can result from punishing and rewarding. For example, punishment can increase stress and aggression and be detrimental to learning (Hodent, 2021, p. 16).

Next is emotion, which is a complex topic and will be discussed in a simplified way regarding what is happening in the brain. Emotion is referred to as a state of physiological arousal, for example, an accelerating heart rate and sweaty palms. In cases of fight-or-flight, one's body might start producing hormones (adrenaline and cortisol) for the purpose of raising awareness and tensing one's muscles. Another part of the brain, the amygdala and hippocampus, will identify the situation, compare it to previous memories, and store the new experiences so they can be used in the future to avoid danger (Hodent, 2021, p. 18–19).

"*They frequently say that people who play World of Warcraft have an addiction problem and that they are wasting their lives, not realizing that their own grinding for achievements at times bear a striking resemblance to this type of gameplay.*"

Jakobsson (2011, p. 8)

There are also addictive aspects of achievements, as highlighted by the quote above. Jakobsson (2011, p. 8) observed that this behavior was particularly evident among *World of Warcraft* players, likely due to the game's repetitive and reward-focused design. There were also cases, not specifically in *World of Warcraft* but on Xbox 360, where people offered their services for a fee to boost other people's Gamerscore points.

While achievements provide motivation and reward, they share structural similarities with addictive mechanics. To go further into this and to understand it, we will look at loss aversion. The core concept of loss aversion is that gaining something makes you feel good, losing something makes you feel bad, and losing something makes you feel worse than gaining the same thing makes you feel good. This is more about losing something in the game, whether tangible or intangible, and not about losing the game itself (Engelstein, 2020, p. 4, 6; Hodent, 2021, p. 19). What Engelstein (2020, p. 11) points out though is a key difference between achievements and other addictive gameplay: once earned, they are a permanent reward and do not need to be repeated, but there have been cases where unobtainable achievements have been shown and have never been able to be obtained.

2.4 Achievements as a reward

The word "reward" has been used a multitude of times throughout this research, therefore it becomes important to go into what it means for this research. Humans have a fundamental need to be judged, a deep inner need to know where we stand in the world, but what we crave is to be judged in a favorable direction, and rewards are a way the game tells the player, "you did good" (Schell, 2010, p. 162, 231).

Rewards can look like:

1. **Praise** – The game tells you, through, for example, explicit statements, a special sound effect, or an in-game character speaking to you, that you did well.

2. **Point system** – A way to measure your worth. Sometimes these points are also an approach to another reward system, but can also just be shown off, for example, in a high score list.

3. **Prolonged play** – Making you, as a player, have more time to play the game through rewarded resources, more time to play, or more life to get a higher score or measure of success. It taps into our need for survival.

4. **Gateway** – Plays with the player's desire for exploration and is the reward when the game shows you more areas, for example, unlocking more levels or receiving a key to a locked area.

5. **Spectacle** – Being rewarded with beautiful and interesting things, usually in the form of music or animation. This is seldom enough to satisfy a player and is usually used in combination with other rewards.

6. **Expression** – Letting players express themselves through special clothes or decorations. This satisfies a basic urge for the player to look different from the rest in the game but might not satisfy the overall goal in the game.

7. **Powers** – Reaching a more powerful state in a game is something anyone might desire in real life. In a game, it is likely to improve the game's judgment of a player's success. Powers allow the player to reach the game's goals more quickly than before.

8. **Resources** – Games usually carry resources, such as food, energy, ammunition, or, in some cases, virtual money. Games with virtual money give the player the choice of how to spend it, for example, on other resources, powers, prolonged play, or expression. Free-to-play games that use real money can blur this line by allowing conversion of real money to virtual money or resources.

9. **Status** – The player's status can be a very desirable reward, especially to competitive players. Examples include high leaderboard rankings and special achievements.

10. **Completion** – Completing all goals in a game can give closure to the player, something hard to compare to problems in real life. After the completion of a game, there is often no point in playing further (Schell, 2010, p. 231–233).

What happens with achievements, aside from being goals with rewards, is that they may also satisfy one's desire to collect or one's need for closure. By completing all the achievements, one might feel a sense of closure over the completion (Groening & Binnewies, 2019, p. 12).

2.5 The amount of achievements

Getting rewarded constantly can lead to becoming acclimated to it, and what was special before might start feeling like nothing special anymore. To overcome this, developers tend to increase the rewards later in the game, as well as remove the predictability of rewards. An example of predictability is monster drops, where, to remove the boredom and predictability of getting the drops, there is instead a 1/3 chance of getting the drop, making it more of a surprise to receive (Schell, 2010, p. 233). This is something achievements lack the ability to do, and there is no clear way to solve the issue of becoming acclimated, aside from starting a totally new game.

Groening & Binnewies (2019, p. 11) mention achievements are effective in enhancing performance and motivation. When it comes to motivation and the quantity of achievements, research seems to indicate that at least a low amount of achievements leads to task persistence compared to no achievements, while a high amount of achievements does not. Getting acclimated to achievements was one of their hypotheses as well, and it might bring a value of a few rewards being higher when one has to work longer and harder to achieve. There is also research showing that a low quantity of achievements leads to a higher perceived achievement difficulty than a high quantity of achievements, resulting in the perception that higher amounts of achievements are easier to achieve.

When comparing two different tests with different difficulties of achievements, the group with difficult achievements outperformed the group with easy achievements. However, while they did this, they did not adjust their performance to the difficult achievements, but instead to the easier achievements in their respective group, assuming the group reached their natural performance limit and that the difficult ones were just too difficult. While this shows that players perform better with achievements, and definitely with difficult achievements, there is little research that backs it up that

they enjoyed their tasks or showed more interest in the specific task that gives the achievement (Groening & Binnewies, 2019, p. 11).

2.6 Dark game design pattern

At this point, we have discussed the power that achievements can have over players based on their motivation and feelings. They can be used to present rewards in front of the player to motivate them to complete tasks or activities in games. This does not specifically have to be bad, but when involved with dark game patterns, we can end up with cases where the player is subverted from what is going on and unable to make autonomous and informed choices.

""dark patterns" - describing instances where design choices subvert, impair, or distort the ability of a user to make autonomous and informed choices in relation to digital systems regardless of the designer's intent."

Gray et al. (2024, p. 2)

It boils down to (1) a negative experience for the player and (2) the intention, on the part of the creators of the game, to cause that negative experience. These do not happen by accident and are fulfilled when they are purposely evoked; they can also be evoked later, even after the game. These are divided into categories of time, money, and social capital that the player could be deceived into spending or using (Zagal, Björk, & Lewis, 2013, p. 2-3). Achievements are not inherently manipulative should be prefranched again, but for people that have an interest in achievement it is a system that works on giving the player feedback, goals and validation that in the next sections can become problematic when they are designed to steer these players into playtime, excessive spending and social obligations.

Robbing a player of time can be broken down into (1) whether a player can develop a sense of the time commitment necessary to successfully play the game, (2) whether the player's expectations of the time commitment are significantly at odds with the actual time required, and (3) how likely players are to feel they "wasted their time" (Zagal, Björk, & Lewis, 2013, p. 3). This can also be called social engineering or engaging strategies, making the person keep using the media for as long as possible by entertaining them and occupying them on the subject (Gray et al., 2024, p. 13).

The situations mostly used here in combination with rewards are "grinding," where a player needs to spend a long time in a game repeatedly performing the same action to get something out of it. This can also take advantage of a player's competitive side, where beginners need to put in an extra amount of time in a game to avoid losing to a more experienced player, in which case grinding is used to protect their enjoyment of the game. There are also cases where a game asks you to log in at specific times to collect or receive something in-game, or where something in the game expires if you do not repeatedly take care of it (Zagal, Björk, & Lewis, 2013, p. 3–4). As an achievement this can look like having to collect all seasonal items making the log in at specific times important or an achievement that affects the grindy aspect for example collecting materials that are randomized to what you get, essentially achievements that focus on adding more hours to be able to fulfill. This can also potentially relate to Fear of Missing out (FOMO), a phenomenon about the uneasy feeling that others might be having a rewarding experience from which one is absent from. This has also become a commercial tactic to convince people to buy or interact through advertising (Hodkinson, 2019, pp. 1,3) for example as specific login times in a game.

Moving over to dark patterns involving money, we have (1) the likelihood of a player regretting having spent money to play the game, (2) the player "losing track" of how much money they are spending while playing the game, (3) whether they are aware of what they are getting in return for their money, (4) the likelihood that the player will inadvertently spend money in the game, and (5) whether they are aware of how much money they will have to spend to achieve their goals in the game (Zagal, Björk, & Lewis, 2013, p. 4). Games sometimes navigate the user toward decisions that the platform providers prefer the user to click or interact with, this being called obstruction or a governing strategy (Gray et al., 2024, p. 13). Just as with time, this can also exploit players' competitiveness, with options for new players to buy their way up to a comfortable level of skills and items to protect their enjoyment of the game, a "Pay to Win" option (Zagal, Björk, & Lewis, 2013, p. 5). For players a "pay to win" can help speed up the process to get achievements that otherwise could have taken much longer to get, this could also for example being an achievement tied with a character or item that cost money to get in the game.

The last category in dark patterns is social capital. Its purpose in socializing is reflected in (1) whether a player's social standing can be diminished as a result of playing the game and (2) whether it is likely the player will feel that they must play primarily because of a sense of social obligation. Some games require the player to have "friends" in the game to make progress or unlock things. This can lead to persuasion to make people join the game if they do not already have friends in it. The inviting part is not the problem here, but the entrapment that players experience, feeling socially obliged to play and continuing the pattern of inviting more people (Zagal, Björk, & Lewis, 2013, p. 5). There are achievements out there that can be about adding friends in the game, or get enough likes and comments on a game's social feed to get.

2.7 The social dynamic

With achievements there is also a social layer, such as comparing achievements with friends, adding another layer of motivation, and can spark friendly competition, especially for achievements that are known to be time-consuming (Engelstein, 2020, p. 11; Jakobsson, 2011, p. 9).

Usually, game platforms store information about what achievements are completed. When receiving one, there is also a cue that the completion happened. Through these game platforms, they are usually visible to other players as well and can be seen, this being the case for Steam. This can also look in the form of points, for example in Xbox Live, where the achievements are boiled down to the number of points that are rewarded and are called Gamerscore (Hamari & Eranti, 2011, p. 11).

For more difficult achievements that involve multiple people collaborating, there are activities organized by groups to obtain them together through websites. Boosting is the word used to describe playing the game specifically to achieve the reward that requires cooperation to achieve. There are also cases of friendly competition with friends, for example on Xbox 360 they are presented as scores, with the winner being the one with the highest score (Jakobsson, 2011, pp. 8–9). While it can be fun and prideful to be able to let others view one's profile and game history, there are also cases where this instead turns into a chore of keeping up the lifestyle (Jakobsson, 2011, pp. 10–11).

There are cases where one's game profiles, specifically gametags on Xbox 360, have had a large number of views and investigations, when revealed to the public. Game critics are judged for not having achievements in a game because of the belief that they have not played the game, even if this is something that could have been played on a separate system that does not activate achievements.

As well as famous people being critiqued for whether the information they say is true or not when talking about if a game has been played (Jakobsson, 2011, pp. 10–11).

Seeing an achievement through someone else's platform can motivate a player to pursue finding out more about the achievement themselves, in some cases trying to emulate the person who has been rewarded by doing it themselves. When it comes to incentive motivators, having social recognition has been proven to show a positive effect on task performance. Achievements that tend to be more competitive have been shown to be important in video game enjoyment and can at times act as resumes when finding parties to play with (Blair, 2011, p. 26).

2.8 How is an achievement designed

A doctoral dissertation by Blair (2011) goes over different reasons a player might find achievement at the locations they are placed in the game and its game design purpose. Blair (2011) provides a useful conceptual framework for examining how extrinsic reward systems may interact with player motivation. Providing points of interest for the empirical interviews to evaluate.

Timing of achievement delivery is all about when the player receives the reward, whether it be straight away or later in the game. Getting the achievement later can be crucial in more intense scenarios where getting the achievements straight away can break the immersion or flow for the player (Blair, 2011, p. 14-15; Blair et al, 2016, p.2). “Theory of flow” is a deep concentration and feeling immersed in the activity at hand created by Csikszentmihalyi (1991), something very common for game developers wanting their players to feel when playing their game. A feeling that is described as being the secret to happiness, for its optimal experience. A person's body or mind is stretched to its limits in a voluntary effort of accomplishing something difficult and worthwhile (Hodent, 2021, p.18). Breaking this flow, through for example an achievement notification in the middle of the game could break this immersion of motivation, control and enjoyment that you get from flow. Getting an unexpected notification could make players' attention go towards instead if they are doing well in the game or question if they are doing what they were supposed to be doing. A way to combat the unexpected notification could be to have it play after the game match or activity is finished in the game. An achievement instead being received when it is expected and when purposely going for it could enhance the flow up to the point where you receive the achievement. (Blair, 2011, p. 14-15; Blair et al., 2016, p. 2).

This to not be confused with *expected and unexpected achievements.* While the notification can disturb or help performance, it can still hold value in receiving a reward when unexpected. A player either knows when to expect the reward or they do not know when to expect it. Unexpected entails typically being still aware that the game has achievements, but not knowing what will cause the reward. There is positive to both, while expected achievement might provide the player with opportunities to set out goals for themselves, this also having the potential of motivating learning in a game. Unexpected ones might give incentive to experiment and explore outside of normal play, but might not carry the ability of learning in the same way that boosts performance such as expected might do (Blair, 2011, p. 13-14).

There are a couple of achievements that work with quantity and game grinding such as *incremental and meta achievements*. An example of this could look like catching fish in a game, where the player receives one at different incrementals like 25, 50, 100 and so on. Completing a series of a bundle of similar achievements and receiving one for that, like catching all the fish in the incremental example would be called meta achievement. This can be used in a way to break up a player's progress into

more specific goals and grouping them together will let the player know they are connected and completed. A lack of interest from the player can be shown if they feel that the achievements are perceived as little more than a reward-driven incentive structure, rather than activities that provide meaningful challenge or self-directed engagement (Blair, 2011, p. 20-21 ; Blair et al, 2016, p. 3). Another word used by Hamari and Erantti (2011, p. 10) in exchange for incremental for similar purposes is *multiplier*, an action in an achievement that has to be carried out repeatedly.

Goal orientation is about what the players goals are to the situation, this is typically followed up with performance orientation or learning orientation in the players goals. The performance orientation is “seek to gain favorable judgments of their competence or avoid negative judgments”, while learning is "individuals seek to increase their competence”. To clarify what this might look like is someone with a high performance orientation might take fewer risks and experience less when participating because they might feel a fear of failure for different reasons, example being affecting their score in a game. Someone more aligned towards learning orientation might accept failure and might seek challenges more to provide opportunities to learn. Achievements that lean more towards performance orientation players have a tendency to perform better on simple tasks while goals that are more towards the learning orientation perform better on complex tasks (Blair, 2011; Blair et al, 2016).

Competitive and non-competitive cooperative achievements look at the social aspects of achievements (Blair, 2011, p. 21-23). The competitive ones pit players against each other and this can be done through either direct confrontations or indirectly through other interactions for example scores or tasks, while the non-competitive ones are based on two or more people working together, working together with a peer, mentor or protégé.

Competitive environments have some positiveness to it that it can increase intrinsic motivation by influencing one's competence, it has also shown to increase players performance on easy tasks. Instead a more overwhelming amount of studies have shown that it interferes with the learning process in the form of that it encourages egocentric behavior and can add negative effects on players' self-efficiency through deteriorating feelings of self determination (Blair, 2011, p. 21-22).

Non-competitive cooperative games have shown an improvement in performance in comparison to individual incentive programs. These kinds of achievements can also boost the encouragement of more advanced players assisting less experienced players to reach goals that would be too complex or locked from the player when playing alone. A downside to this could be a group going for more risky decisions or more cautious moves instead of the players preferred action, this being called “group-induced attitude polarization”. The opposite can also happen where individual performance can bring down the rest of the team through for example putting in less effort (Blair, 2011, p. 22-23).

Achievements tend to be received for positive performance in a game, but achievement can also be given for poor performance. Examples here are achievements when you are losing fights or ranked low in a game (Blair, 2011, p. 14). This aligns well with Schell (2010, p. 234-236) on punishment, where adding punishment can increase value in one's choices, the risks have consequences and adds a layer of challenge to overcome. While this can increase enjoyment, it is shown in studies that whenever encouraging the players is essential it is better to reward than punish. Blair (2011, p. 15) pushes on this that negative achievements can decrease one's sense of competence and self determination, as long as the goal in itself is not to pursue the achievement on one's own will.

When looking at how difficult an achievement should be, there are different ways something could be difficult, through for example different gameplay. One easier to measure is that if something is too

difficult a player will not attempt it and if something is too easy it will be completed too quickly and will not provide adequate challenge. It becomes valuable to match it to the player's skill, so they view it as challenging but not critically difficult in the sake of difficulty (Blair, 2011, p. 5).

Measurement achievements look at the players performance and measure it, these can look less like the completion achievements that are given for completing a task and can be more like beating a high score both ones own or in a community or for example if a game has a star rating system then it can be being able to beat the same task to different degrees of difficulty. These can be seen as a type of feedback for performance, the feedback to reflect on one's performance in the game (Blair, 2011, p. 18).

Task-contingent rewards, completion being the goal are what achievements are best described as and what a major part of the research falls under. These types of achievement are either seen as completed or not. There are performance contingent ones that require skill to achieve and non-performance contingent are usually received without the need of skill or ability and might be more a reward for attending a game event (Blair, 2011, p. 19-20). What Hamari and Eranti (2011, p. 13) view as completion is a trigger a player invokes or that system invoked, and follows on how many times it has been triggered, under which conditions and what pre-requirements.

Some achievements need the right gameplay options to achieve, they have pre-requirments to them. This could be the need to have the right game mode, difficulty or character class picked before the gameloop starts, it can also be about playing at the right time or right season (Hamari & Eranti, 2011, p. 9).

Design Factor	**Description**	**Effect on player**
Timing of achievement delivery	When the player receives the reward, whether it be straight away or later in the game.	Can break the immersion or flow
Expected and unexpected achievements	A player either knows when to expect the reward or they do not know when to expect it.	Expected achievement might provide a set out goal. Unexpected might give incentive to experiment and explore outside of normal play.
Incremental and meta achievements	Receives one at different incrementals like 25, 50, 100. Meta, completing a series of a bundle of similar achievements.	Can be used in a way to break up a player's progress into more specific goals and group them together.
Goal orientation	Achievement designed for performance or learning.	Performance orientation might take fewer risks and experience less when participating. Learning

		orientation might accept failure and might seek challenges more to provide opportunities to learn.
Competitive	Pit players against each other.	Can increase competence and motivation but may harm learning.
Non-competitive	Two or more people working together.	Encourages teamwork and mentoring.
Positive vs. Negative	Rewards success or poor performance.	Can encourage behavior or reduce feelings of competence through increasing the importance of one's choice.
Difficulty	The challenge level of achievement.	Valuable to match it to the player's skill, so they view it as challenging but not critically difficult.
Measurement achievements	Measures players performance, ex. High score.	Feedback to reflect on one's performance in the game.
Task-contingent rewards	Earned by completing tasks.	Rewards the player for being active, without any need of skill or ability.
Pre-requirments	Specific requirements before achievement becomes available.	Supports different gameplay or replayability.

Table 1: Summary of design factors

An article from De Sales and Lewis (2013) brings up some different types of achievements that might appear in games. They do this with both looking at previous research bringing up what they might be lacking from older sources but also going over 49 games from 12 different genres, as well as ending up being 7 different platforms. This ends up being 1202 achievements categorized into 33 different groups for achievement. There are some years now to this source so that 33 might not be accurate nowadays, but this shows there are at least many different directions a developer may go with designing them. For more specifically on what 33 ended up being and the comparison to previous research the authors did, I advise reading the article oneself.

2.9 When achievements gives the player more than just “a pat on the back”

There are times when the achievements system is tied to more stuff in the game than just a reward that appears on the game platform. Sometimes it is tied to virtual goods or artifacts in the game, for example the early case of achievement the patch previously mentioned (Jakobsson (2011, p. 5), or in Team Fortress 2 (2007), it awards the player with new weapons that you can use in the game when completing specific achievements. Team Fortress 2 (2007) is in a very short summary a multiplayer shooting game, where the items that are purely aesthetic will not affect the gameplay, but has shown to bring a reputation and competition through a social system (Hamari & Eranti, 2011, p. 11). Sometimes the digital rewards show up in multiplayer settings and the motivation here might fall more under a social reflection of one's self, either in reflection in one's skill or in a way to express ones self (Blair, 2011, p. 16).

Even if the achievements are earned items in a digital setting, there can still be digital value in the form of an item, cosmetic change or title. Receiving an item or anything like this is said to affect one's intrinsic motivation and decreases the likelihood of a player returning to the task after obtaining the reward. The items may get the players to pursue the items instead of pursuing intrinsic motivation (Blair, 2011, p. 16). Ryan & Deci (2000, p. 3) also mentions this where tangible rewards that are task performance focused affect intrinsic motivation, because it becomes a focus on extrinsic effect instead, focusing on the means to an end rather than something persuaded of its own sake. For example with achievements once complete, motivation to continue the activity may decrease because the external incentive is complete.

Some games use achievements as currency, where the achievements accumulate into points that can be used to spend on in-game special items or real world objects. Has been shown to be a boost in one's performance gain when compared to tangible rewards (Blair, 2011, p. 25). For example in World of warcraft, quests can give achievements, where they are a part of the reward to the quest (Jakobsson, 2011, p. 7). Giving a real life example of achievements or other similar game patterns for example patches, there is for example marketing campaigns where users receive loyalty stamps (Hamari & Eranti, 2011, p. 2) something that also works with the concept of collecting.

2.10 Achievements as data and educational tool

From 2004 to 2021 Cunha, Mendes et al. (2024) analyzed Steam achievements in games, they examined for engagement and player persistence using achievements as a progress marker. Their findings landed in that while they were limited with being able to distinguish specific conditions towards other conditions, for example if a game turns off the achievement unlocking in specific game modes, as well as different genres having different play styles. It still showed through a quantitative level that developers prefer to make games that demands a certain intermediate persistence and that a games player base decreases over time through playing (Cunha, Mendes et al., 2024, p. 9-10).

Similarly Bailey and Miyata (2019) examined achievements earned through completing a single player's game’s main scenario to understand when players stop progressing in a story and found that very rarely does a player finish the games they start.

Articles such as Heeter, et al., (2011), Blair (2011) and Blair et al. (2016) about achievements follow a serious game, or at least education approach that achievements can have, based on how they are treated in video games. Heeter, et al., (2011) does a comparison to this in a classroom setting.

"Achievements in school are rewarded by sense of mastery of subject matter, progression through grade levels, honor roll lists, and praise from teachers and peers. Achievements in games are rewarded by a sense of mastery of the game, progression through levels, by points, prizes, inclusion on leader boards, and admiration by other players."

Heeter, et al., (2011, p. 43)

As well as Blair and Blair et al. (2011, p. 2; 2016, p. 48) researching achievements from a serious game standpoint., believing that the feedback and reward that comes from achievements in games can be a beneficial for educational games, and as already mentioned previously it can bring players to increase playtime, that can be beneficial for learning.

3 Problem

Previous research has explored achievements primarily as game mechanics that influence motivation, engagement, and player retention. However, there is limited research focusing specifically on how players themselves perceive achievements and what makes them meaningful. While achievements have become a common feature in modern games, there remains a gap in understanding why certain players invest significant time and effort in obtaining them.

This gap is relevant both from a research and industry perspective. Achievements are often expected by players, and games released without achievement systems frequently receive requests from their communities to implement them (Steam community, 2024). Despite their widespread use, it is still unclear what specific value achievements provide to players and which motivations drive achievement-oriented play.

A better understanding of these motivations could help game developers design achievement systems that support player engagement and satisfaction more effectively. Therefore, this study seeks to answer the following research question:

What motivates players to continue playing a game in order to collect achievements?

3.1 Method

To address the research question the methodological approach consists of a case study and qualitative interviews. Each method is described under its own heading, outlining its specific purpose within the study. However, the rationale for adopting this method will be discussed first. The case study, in this context, functions as an iterative process and primarily serves as foundational work for the remainder of the methodology, see figure 1 for reference. In contrast, the interviews aim to elicit in-depth qualitative responses, particularly focusing on players' emotional experiences. Lastly the thematic analysis brings out what patterns and understandings the experiences from the participants could have and analyzes them.

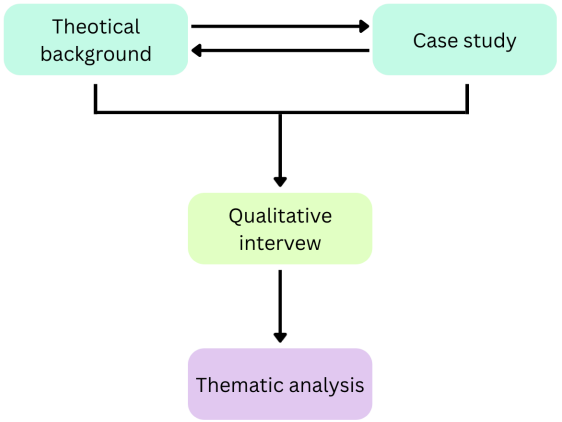


Figure 1: Showing how the information is handled throughout the research

3.2 Case study

Examining how a game platform implements achievements is also important for this study, as it helps explain how players may be motivated to continue playing in pursuit of completing achievements.

This through looking at the design of the interface and how achievements are presented and communicated to players. Studying Steam's achievement system therefore provides insight into how interface design and platform features can encourage players to notice achievements and monitor their progress. This may also be of interest to developers seeking to understand how achievements are communicated across the Steam platform.

The platform Steam, developed and distributed by Valve Corporation, is a digital distribution platform for games, as well as a provider of digital rights management, multiplayer, and social networking services. It is accessible from Microsoft Windows, macOS, and Linux (Wikipedia, 2026).

The focus of this study is the largest game platform at the time of writing, Steam. The research is limited to a single platform due to time constraints and limited familiarity with other platforms. There are, however, a few comparisons that will be made to other platforms, and these are included through existing research in the theoretical background. Specifically, there appears to be limited research on achievements within the Steam platform, based on available sources. Much of the existing literature instead focuses on platforms where achievements are mandatory. This places Steam in an interesting position, as it hosts both games that include achievements and those that do not. The case study therefore provides an opportunity to explore the potential benefits and differences between these approaches, particularly in relation to the platform's interface.

The study will follow a single-case design with multiple units of analysis, the units being cases involving achievements (Säfsten, 2020, p. 108). Because the study concerns a digital platform, the primary method of investigation was observation, supported at times by the theoretical background when relevant and by internet searches when observation of the website was not sufficient.

As the study is limited by the author's own experience and understanding of the selected platform, there is a possibility that certain relevant aspects may be overlooked, the author themself having a background in game design and an interest in achievement. Additionally, the way the platform handles achievements and rewards may evolve over time, meaning that findings could become outdated. This limitation aligns with concerns raised by Bell (2010, p. 9), who notes that case study research can be prone to selective reporting and potential distortion.

The investigation is conducted using the author's personal Steam account. Any unrelated information has been removed from images, and the selected games used to illustrate achievements were chosen for convenience. These choices, however, are not expected to significantly limit the validity of the observations related to achievements, because the subject in itself is still being shown.

3.3 Qualitative interview

Empirical interviews were conducted based on the theoretical background and insights gained from the case study. This method was chosen for its ability to gather information in the form of participants' perceptions and experiences with achievements (Säfsten, 2020, pp. 152–153).

The study employs semi-structured interviews, allowing flexibility to ask follow-up questions and explore emerging topics in greater depth. The approach aligns with the characteristics outlined by Merriam and Tisdell (2016, p. 110). The interviews consist of a mix of more and less structured questions, offering flexibility while still focusing on specific data related to achievements (appendix A). Although responses are explored in depth, all participants are asked a similar set of core questions, without a predetermined wording or strict order.

The sample group was selected using non-probability sampling, specifically criteria-based and purposive sampling, to identify typical and ideal participants. The selection criteria focused on individuals who play games on Steam and are motivated by achievements. This sampling approach is based on Säfsten (2020, pp. 135–137).

The participants selected for the interviews are individuals who experience a motivation to complete all achievements in a game, enabling a deeper exploration of the motivations behind this behavior. There are no restrictions regarding game genres; however, there is a focus on individuals who regularly play on Steam, in line with the case study. Including participants who play different types of games allows for a broader range of perspectives and may support future research into how different game designs influence achievement-oriented behavior.

Participants were recruited through local communities, primarily to facilitate in-person interviews. These communities consist largely of individuals involved in game development, which may influence the findings, as participants might possess a deeper understanding of game design than the general population, which is not a partially negative addition in itself. To ensure transparency, participants' backgrounds such as experience in game development, research, or psychology were noted during the interviews. The interviews were conducted using both voice recording and note-taking. Note-taking was used to highlight particularly interesting points during the interview, while the recordings were transcribed to enable further analysis. The transcriptions together with voice recording were then used to identify themes within the data.

A thematic analysis was conducted following Braun and Clarke (2006). This approach emphasizes identifying patterns within the data (Braun & Clarke, 2006, p. 12). The analysis followed six steps: (1) familiarizing oneself with the data and noting initial ideas, this step got followed up through the notetaking; (2) coding the data and adding annotations, used a digital bulletin board; (3) organizing codes into potential themes, placed the digital points in the bullet board in potential themes; (4) reviewing the themes; (5) defining and naming the themes; and (6) producing the final report. This process also aligns with Merriam and Tisdell (2016, pp. 207–208).

A semi-structured pilot test was conducted to evaluate the quality of the interview design and its ethical considerations. The pilot interview was carried out in the same manner as the main interviews, followed by feedback. Minor adjustments were made to the initial contact with participants; however, no changes were made to the interview questions. Due to the limited amount of these changes, no further pilot testing was deemed necessary.

3.4 Research ethics

Participants were informed about the level of anonymity provided in the study, as well as how the responses would be collected, both being notetaking and voice recording. The participants were informed about how the interviews would be transcribed. When it came to letting the participants know what the interview was about they got a rough outline of what their contribution would give and that it would be summarized and quoted in the result section.

Participants were also informed of their right to withdraw from the study at any point prior to the publication of the paper, knowing as well that it was going to be published. When the interview was conducted they were in person at the University library in a quiet separate room, whenever the interview went into more unsure or sensitive directions, there was reassurance that it is up to the participant on how they want to answer, that they can take their time and also that it is okay to skip questions when feeling so.

To keep the anonymity further demographic questions such as gender, age and playing habits have been summarized in the results section.

3.5 Transcribing the interview

Klang.ai is an AI-based tool that converts voice recordings into text and can also generate summaries of the content. Participants were informed of the use of Klang.ai before the interviews. The transcribing tool ended up being used for reading along with when going over the voice recording for patterns and interesting comments for the thematic analysis, as well as a reference point for locating relevant sections of the interviews more efficiently when revisiting specific topics. Its ability to find patterns in a text and summarize the content was not used for the result section, but was used to navigate around the text for finding where the topic was talked about and then also accompanied with listening to the voice recording.

The participants and interviewer shared at least bilingual languages, so there was a recurring switching between languages in all sessions for participant comfort. Klang.ai had no bigger problems with this, and was tested beforehand with pilot-test, although some inconsistencies in automatic translation were observed such as sometimes not translating.

3.6 Generative AI

Academic peer-reviewed articles were primarily sourced through journals and Google Scholar, but in addition, tools such as ChatGPT and Scholar Labs were used to assist in identifying relevant academic literature on the topic being able to pinpoint more clearly what felt like was missing from the research.

Generative AI was also used for grammatical revision and to improve sentence structure. All content was originally written manually and subsequently reviewed using the AI ChatGTP for language accuracy. This process also applied to the development of interview questions.

As mentioned previously as well Klang.ai was also used, it is built on different kinds of AI, but some of it is generative AI. How this got incorporated and validated for the study has also already been mentioned.

4 Case Study Findings

4.1 Icon, name and description

This section provides a case study on the game platform Steam. As Hamari and Eranti (2011, pp. 5–6) have previously discussed, the visualization of name, icon, and description follows the same pattern on Steam in how they present their games. On Steam, they also show the date of unlocking and the global percentage of people who have unlocked it.

As Hamari and Eranti (2011, p. 7) mention, achievements can be vague in their descriptions, where you need game knowledge to understand them, and sometimes the description only mentions a location and nothing else. Continuing this, achievements can at times deviate from typical game design patterns; for example, with the achievement in Hypnospace Outlaw (2019), "Brassed," where the only description you get is "Tubaculosis!!!" (Figure 2), something that is arguably vague to the point where having game knowledge is not enough to grasp it.



Figure 2: An achievement from the game *Hypnospace Outlaw* (2019).

Another example of an obscure achievement is The Stanley Parable (2013), where you receive an achievement for not playing the game for 5 years in order to unlock it, going out of your way to not be part of the gameplay to achieve it. And let's not start with their "Unachievable" achievement and its description "it is impossible to get this achievement". Exactly as Engelstein (2020, p. 11) mentions, there have been cases where unobtainable achievements have been shown and have never been able to be obtained, but even in these cases, they have been obtained. The guides on how to get this achievement are instead about how to configure one's game (Steam Community, 2017).

Continuing on game guides for achievement, there are also multiple of them out there for games popular enough to have a community showing how to get all the achievements in the game through documentation directions. Making it possible for any players to look up how to complete them.

Developers have a couple of design choices, as discussed, with icons, names, and descriptions. The icon might shift in color when unlocking it, categorize different achievement types or difficulties into different design styles, or simply make the achievement a secret unlockable.

4.2 Secret achievements

Some achievements are chosen by the developer to be hidden from the player by hiding their title, icon, and description. A player, if they want to, can always check through Steam how many achievements there are in a game, but not always see the information on how to obtain them. This can be bypassed by going to the game's achievement menu to view the global list instead of a personal list of achievements, making it possible to see secret achievements when clicking on them.

When a friend receives a hidden achievement, it shows up in the game's activity feed as a hidden achievement and is not viewable. However, by navigating to the activity folder under the player's profile name section, one can view a friend's achievement as if it were a regular achievement.

4.3 Stream overlay and notifications

About what happens when a player gets an achievement in-game, if no settings or notifications are turned off or changed, Steam displays a pop-up window in the lower right-side corner showing which achievement you unlocked, along with its name, icon, and description. There is also a specific sound jingle that is played and remains consistent for each unlocked achievement. This notification also shows up in the player's game profile in their library of games, both under the activity feed and the achievement folder (Figure 3). It shows information aside from the name, icon, and description, including the date the achievement was unlocked, as well as the global percentage of players who have the same achievement.

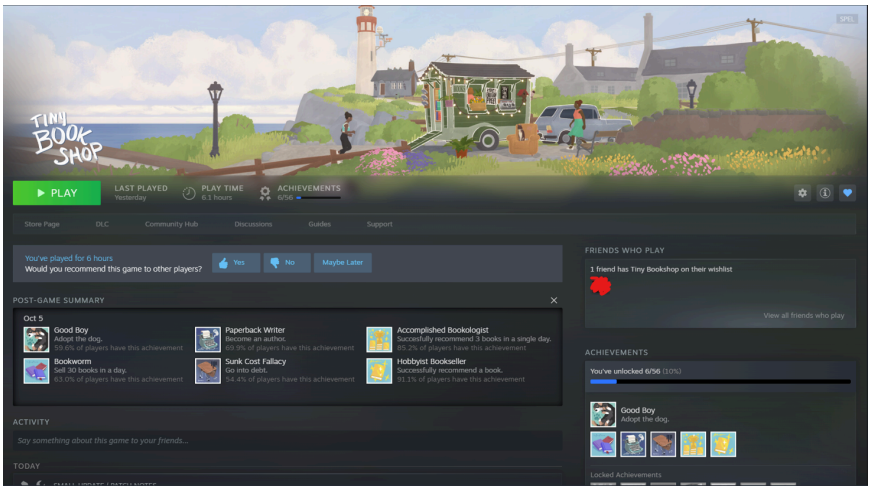


Figure 3: Game profile of *Tiny Bookshop* (2025), showing the post-game summary with newly unlocked achievements and the achievements folder.

To the right of the activity feed, there is a window with more specific information about achievements if a game has achievements. First, there is a progression line showing the number of achievements left, followed by an icon, name, and description of the latest one, and then only icons of other recently received rewards. The next row shows some not yet received. If the game has a larger number of achievements, it shows only a few and obscures the rest of the not yet received ones. If you have fewer achievements than what the row is able to show, there will be empty space in those spots, and when everything is fully complete, the row disappears. If the row with received achievements or the row with not yet received achievements is not able to show them all, there will be a spot with the number of achievements not being shown. If you want to see the rest, you have to click the text mentioning "view my achievements." Secret achievements that are left never pop up in the row. What determines which achievements are shown in the row of not yet received achievements is based on the global percentage of how many players have them, showing the highest percentage first.

The game's activity feed is also a location where friends on Steam can see each other's unlocked rewards. This can even be followed up with the option to "thumb up" the notification in the game's activity feed.

Steam gives you the option to compare achievements with friends who have also played the same game, providing a menu where you see what you and they have left, as well as the time and date you and they unlocked one if it is unlocked. There are two different menus found when trying to compare this information. One of them is accessed by navigating from the game's library to achievements and

then clicking the compare option. Here, there is no information about who is closest to completing all of them through the number of unlocks; to find out, one has to manually count. The other menu can be found by navigating to a friend's profile and their games, finding the game you want to compare, and clicking game stats, then achievements, and then comparing these stats to your own. This one gives the option to see the amount of completion. Based on what is found, there is no way to compare a friend's achievements to other friends, and it is only possible to compare yourself to another.

In the player's Steam library, there is an option to add a filter with different options for what the player wants to find, where one of the options is to filter out games that do not have achievements, but not the opposite (Figure 4).

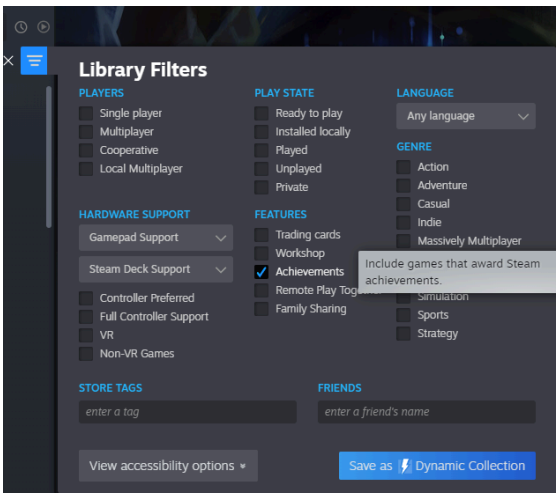


Figure 4: Achievement filter

4.4 Numbers and data

There are achievements that, while not specifically hard to obtain, players usually do not go for. An example is Stardew Valley (2016) with "Joja Co. Member Of The Year". In the game, you are presented with two options: either completing a community center or helping a supermarket called Joja Mart expand in the town. It shares a similar difficulty level with the community center achievement, but even so, there is a clear favorite route for players to pick when playing the game based on the percentage. Going over Cunha, Mendes et al. (2024) or Bailey and Miyata (2019), who researched achievements as data points, this is something that can also be relevant data for a developer to learn which path players prefer.

Every year, Steam releases a game overview called Steam Replay, showing what games the player played and some other statistics, one of them being an overview of the number of achievements and the median for that year (Figure 5). This is something you can later revisit for each year to look at again.

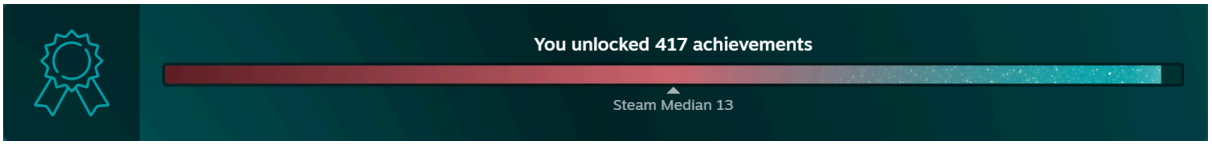


Figure 5: Achievement median displayed in Steam Replay for 2024.

4.5 Customization

There are also customization elements players can add to their Steam profile, where a few of them are tied to achievements in some way. Steam calls these "featured showcases," and the ones tied to rewards are: (1) Rarest Achievement (Figure 6), which showcases players' rarest achievements, the number of perfect games, and the average game completion rate. When hovering over the achievement icon, it also shows the global percentage for that specific one. (2) One called Completionist highlights the games in which all achievements have been earned and the number of completed achievements. There is also an option to hide games from one's profile so they do not appear in the showcases. You do this by navigating to the game in question, going to manage the game, and clicking "mark as private." This makes it private to other people, but not to yourself. This was discovered through the Steam community (2025) and also double-checked by oneself.

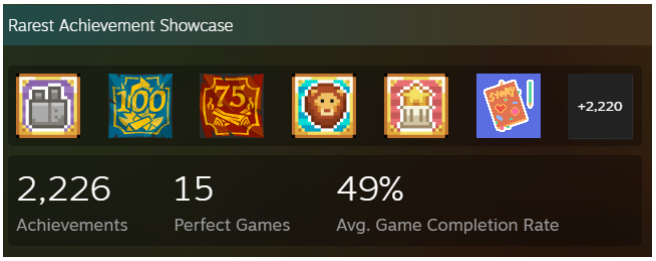


Figure 6: List of the rarest achievements a player has collected, showing the number of achievements, the number of fully completed games, and the average completion rate of games in the player's library.

For a player to be able to customize their profile with showcases, their Steam profile has to be leveled up. The number of showcases you can have is tied to the Steam level. This system is based on a rough estimate of how much someone has invested in their profile, accumulated through, for example, items, badges, and actions that introduce the Steam platform, which give points to level up. There is also the option to upgrade existing showcases by using a currency Steam introduced to purchase customization items, called Steam Points.

4.6 Completion

When the global percentage is lower than 10 percent, it is also presented with a shine effect around the achievement. As well, when you get all the achievements for a game, the folder is presented with a ribbon, signaling that it is complete (Figure 7). This aligns well with Schell (2010, pp. 231–233) regarding the spectacle reward, where the player gets rewarded with special effects.



Figure 7: Achievements folder in *Tunic* (2022), showing all achievements completed, including golden particles for achievements with a low global completion rate.

In a player's profile, when navigating to where the games are, there is a folder for perfect games showing all the games where one has completed all achievements, together with a number indicating how many are in the list. If a game does not have achievements, it does not show up here and is solely for games that have the reward mechanic.

For games with downloadable content (DLC), they sometimes come with their own achievements, and sometimes they are added to the main game's list. This can mean that achievements can sometimes be placed behind paid DLC. To 100% a game, you might also need to buy all the content.

4.7 Miscellaneous

A game that does not have achievements does not look that different from other games on the Steam platform and more closely just misses the features that accompany games that have achievements, the features being the ones previously mentioned in this section. The lack of content means fewer features are incorporated into the game platform, but as Hodent (2021, pp. 6–7) says, the brain perceives information that we process through attention, motivation, and emotions to form memories. This can form a norm about what to expect that might not feel met when achievements are not an option and someone is used to that kind of gameplay option.

Steam has maintenance windows when it affects all kinds of services on Steam, achievements falling under services. Services can also go down for other reasons, such as DDoS attacks or high traffic on the website. Steam, in this case, goes into an offline mode, allowing players to still play games but not reach the internet or connected aspects from Steam, so unlocking an achievement here can mean that you still will not receive it in offline mode (Steam Dashboard, n.d.). Steam, in their guide for developers under the section "offline mode" on how to create achievements for their games, says that the offline state should resolve itself after leaving offline mode (Steamworks documentation, n.d.).

5 Thematic analysis

5.1 Participants

Eight people got interviewed, where the participants were 3 women (37.5%), 3 men (37.5%) and 2 non-binary (25%). The participants were between 20-29 years old, landing on the mean of 23.71. All the participants also play video games daily.

	Most played game genre	**Played game platforms excl. Steam**	**Background that might influence**
Participant 1 (P1)	Indie, narrative, exploration games	Epic games	Psychology courses, game design student
Participant 2 (P2)	RPG, co-op games	Playstation, Itch.io	Game graphics student
Participant 3 (P3)	RPG, action, cozy and puzzle games	GOG, Epic games,	Game animation student
Participant 4 (P4)	MMos, FPS	Mobile games, Epic games, Blizzard	Makes game on spare time
Participant 5 (P5)	Co-op, narrative games	Epic games, EA, Ubersoft Connect	Game design student
Participant 6 (P6)	RPG, life simulation, mystery (puzzle) games	Itch.io, Nintendo switch	Animation student, master game development, game developer
Participant 7 (P7)	Rougelikes/lites , idle and open world RPGs games	Nintendo switch	Gamegraphic student
Participant 8 (P8)	FPS, adventure, cozy games	Battle.net, Nintendo 3DS	3D game graphic student, 3D game graphic developer

Table 2: List of participants and what genre they most commonly play, played platforms aside from steam and background that might influence the interview

5.2 The link between enjoyment and achievements

Something that was the same for every participant was that a game had to be engaging for them to pursue achievements. It was not until they viewed the game as engaging that they felt the need to go for achievements.

The journey for the participants to become aware of the achievements once liking the game was instead different. The early bird, P7, looked at them before starting to play the game to get a feel for what kind of game it would be and how achievements would be incorporated into it, still with the idea that this would probably be a game they liked. This was usually only a skim through, followed by a more proper look later after playing the game for a bit. Most commonly, participants P2, P3, P5, and P6 looked at them in the middle of the first playthrough after figuring out if the game was one they would stick with, and a playthrough could range from a short game to a longer game. P3 usually ended up looking things up early in the game because they could feel bad if they missed something that they could not go back to after progressing in the game. This also applied to the secret achievements, as they did not mind being spoiled by the achievement names and descriptions. they would rather not miss anything in the game than worry about being spoiled. P5 was similar in this regard, also looking early in the gameplay at the secret ones as well, but even if they looked at them, they mostly pursued the achievements during the second playthrough because they felt it could take away from the fun of the game during the first round. After a bit of gameplay, P6 would also get a feel for how long each achievement would take and what skill was required based on the game they were playing, and if it felt manageable, they would start looking into what needed to be done to complete it. For the secret ones, it depended on the game length; if it was a short game, they would wait until after the first playthrough before looking at them. P8 usually looked at them after a first playthrough and started pondering whether a second playthrough was worth the time.

“Excited. It´s a sense of accomplishment. Almost like a smaller version of getting praised for a job well done.”

Participant 2

The emotions the participants experienced when unlocking an achievement were, in most cases, positive, with words used such as “feeling productive” (P5), “excited” (P2, P5), “feeling happy” (P3), “feeling accomplished” (P6), and “feeling good” (P7). For P2, it was a smaller version of getting praised for a job well done. With that there is a little popup and a sound that plays accompanying it makes me like it more, P3.

“It's a nice little goal. And then you get a pling. You get a number goes up.”

Participant 5

For P5, there was also excitement for the achievements not yet completed because they knew they would soon get them as well. P7 pointed out that the game itself usually already gave the player positive feedback, but the achievements added an extra feeling on top of that because they were similar to a checklist in terms of motivation, as it was something they could check off, similar to a checklist of chores. Opposite thoughts came from P3, who pursued achievements first of all because they liked the game, but also because they did not feel as if it was a chore.

"I don´t think I have ever felt excited to chase after a achievement. I have felt excited after I have completed it."

Participant 4

For P4, the excitement was an afterthought once the achievement was completed, and the achievements were more seen as a bonus thing they could do in the game.

5.3 Not unlocking

For P6, there are times when playing a game where they experience moments in which they expect an achievement to pop up from doing something unusual or skillful, or at least feel that they should have gotten some kind of reward. At these times, it almost becomes disappointing when they do not get something, where an achievement would have been a good reward to include.

P3 and P5 also have stories of bugged achievements leading to frustration, where achievements should have been unlocked but, for different reasons, were not. P3 had stories of Steam being down, and because they got the achievement at that time, the achievement did not appear after Steam was back up again. P5 had a multiplayer bug where the achievement only appeared for the host hosting the multiplayer game, while the other people joining were unable to get it.

While achievement guides are common and at times relied on by players, there are times when games have no guides for their gameplay and achievements. This was the case for P3, where a less popular game had no guide, making it nearly impossible for them to get the missing achievements and giving them no way to know where to start looking in the game to find them.

When P7 goes for achievements they have not yet unlocked and feel are too difficult to pursue at the moment, they might feel "that I am too bad at it." They describe it as a looming feeling that washes over them because of unfinished achievements.

There are games out there that have achievements that are nearly impossible. For P5, an example they brought up was a game where the achievement required playing the game with the game developers to achieve something close to impossible for them, making them less interested in the game and its achievements. P2 also had an example that was close to impossible because fewer people play the game nowadays, making the achievement nearly unattainable. In this example, the player had to upload things they created in the game to the internet and receive enough downloads or endorsements from other players to unlock the achievement.

5.4 Replayability

Participant 4 recently returned to an MMO they had previously played after learning that achievements had been added, which made them play more of it. In this case, the achievements did not show what stage they were at in the game, making them feel an urge to ensure that the achievements reflected where they actually were in the game.

"I felt like having such a big part of the game always on the screen. It always shows "Oh, you haven´t done this yet". I felt It was such a incomplete thing on the side."

“So I felt I had to get rid of it. So I just wanted to do as much as possible to feel like it had caught up to what stage I was actually in.”

Participant 4

For P6, looking at the achievement list of previously played games can spark an interest in replaying them. They said that they usually look at the secret achievements because they tend to reveal things they did not previously know about the game, which makes them excited to return to it. P8 said similar things, explaining that if a game has interesting achievements, that can add extra motivation to replay it after finishing it. If they are not interesting, however, it might not spark the interest to play the game again for a while.

It is not always the case that a player completes a game or returns to replay it. P2 said they have never consciously dropped a game they were interested in, but instead that outside influences affect them. This can, for example, be friends wanting to play another kind of game or finding another game that seems more interesting at the time.

“It's basically the whole, I like my sparkly rock, but if there comes along a sparklier rock, my brain wants the sparklier rock.”

Participant 2

P7 does not give up on their games. If there is a skill level they do not yet have to be able to get all the achievements, they might instead leave the achievement for a second playthrough, hoping that during the second playthrough it will feel less difficult and easier to complete as their skill level improves.

5.5 Changing the way to play

“You get a new type of game out of it, from the same game. It’s a bit fun.”

Participant 8

P8 feels like achievements can at times provide new experiences. They can require the player to approach the game in a different way or play at a different difficulty level, making them a good reason to replay the game. P6 thinks that achievements tend to take players down routes they might not have considered before in the game. P5 feels that achievements become a break from the game and its story, changing the route from what is normally expected.

Sometimes it can involve using different combat combinations from what the player would normally choose, or in games where players select personality traits for characters, drastically changing the way one approaches the game, according to P2. For P7, achievements sometimes involve playing other characters, and once those achievements are completed, they return to the main character they usually play. Another example from P3 is playing through the game as an “evil” character or siding with characters they normally would not side with.

For P4, depending on the game, achievements can influence the way they play more in some genres than others. For example, in sandbox games, achievements can have a greater influence because they help them decide what to do in the game. In other games, they might still slightly adjust their playstyle to match what an achievement requires in order to play in a certain way. They find it more fun when achievements support trying something different or encourage playing in a different way.

5.6 Frustration

There were a couple of words that were mentioned more than others during the interviews, such as "depends on game", when referred to achievements acting differently in context to what game it comes with, “feeling special” when getting an achievement or completing all achievements, but one word that stuck out and was mentioned frequently was frustration, or synonyms for frustration for example tedious. Frustration was used both to describe situations involving achievements participants did not want to pursue and the frustration that came before the positive feeling of unlocking an achievement they had been trying to complete for a long time.

P1 does not go for achievements that become remotely frustrating, feeling that those achievements can become too difficult. P5 felt that there is a limit to how much frustration or tediousness they are willing to tolerate, and when these situations become too much, it is usually because the achievements require a skill they know they do not have or do not feel capable of achieving.

“Then there are games one really likes. If it is bad achievements, but one wants them, then it becomes quite, one becomes frustrated by it. One almost becomes a little forced to get through these annoying achievements, because one wants all in the game.”

Participant 8

P8 felt conflicted because they still needed to pursue these more frustrating achievements, which brought no enjoyment, in order to complete all the achievements in a game.

For P2, P4, P5, and P7, the most rewarding or memorable achievements came from frustrating experiences. Completing them became the most rewarding or relieving moments because of the frustration and the journey required to get there. In this case, P4 acknowledged that they could have stopped at any time. At times, it also became a display of skill and pushing beyond personal limits. P7 explained how they were constantly only pixels away from completing a skill-based and somewhat luck-based achievement, becoming frustrated by it but very excited once it was finally completed.

“I was like I am finally done, I can go to sleep.”

Participant 5

For P5, this was a time-consuming achievement that required spending a long time walking around the game world collecting items. The guide being used to locate the items also failed to show all of them, adding another layer of frustration. A reason why this experience became both rewarding and frustrating was its effect on the participant’s personal life, as they had exams the following day. Once completed, they felt they could move on with the game, receive the reward, and continue with their life.

“To me achievements can't make up for the lack of entertainment. But they enhance the reward seeking.”

Participant 2

P2 and P8 both discussed games that are too difficult, bringing up the soulslike genre. These games challenged them to the point where they either no longer enjoyed the game or felt exhausted after playing. For P2, at that point there was no value in achievements when the game itself was not rewarding enough.

For P3, the most frustrating aspect of games and achievements is in story-based games where it is easy to miss something that is required for an achievement, making replaying the game the only way to obtain it, which they considered tedious. This could happen, for example, if an area became inaccessible after progressing too far in the game. There was also a sense of tediousness when replayability was not easy to manage, such as not being able to influence how quickly they could progress through the game. An example would be not being able to load different chapters in games that are chapter-based. It became important for P3 that it was accessible to obtain all achievements, and when they missed something in the game that could not be acquired later, it could create a negative perception of both the game and themselves, as they became too focused on having missed something instead of enjoying the game.

5.7 The challenge

"I try to 100% the games that I really like. But there are a few games that I really like that I know I can't get 100% in. Usually because the skill, like the ceiling for how good you can be at the game tends to be really high, which means that the achievements are really hard to get."

Participant 6

P6 keeps the achievements in mind that seem unreachable in case the opportunity would present itself, but for some, for example achievements that are time-sensitive, they do not even attempt them because they go against how they usually play games. P1 does not remotely go for achievements they view as difficult, finding them frustrating. P5 usually does not engage with difficult achievements or those requiring specific skills because they feel they do not have the energy for them. For P3, it depends on how fun and easy they are to obtain; endings in games are considered fun to get as long as they have a guide and does not miss anything, or if the guide is incomplete.

"So it depends, if it is really such an achievement that needs much more skill, then I can wait until a second play through."

Participant 7

Many of the participants use the word "skill" to explain their ability to complete something in the game or not. P7 sometimes feels like the level of skill required does not always match what is needed, that they are "too bad at it" at the moment, and instead continue in the game hoping that later or in a second playthrough their ability will be high enough to complete what they are stuck on.

"Some achievements are obviously like complete the game on a higher difficulty or do x and y but a lot of them can also be things that don't directly relate to game difficulty but is a different type of challenge. Make sure to be methodical in your playthroughs or discover secrets and explore that kind of thing. It feels like a different type of challenge than just pulling the difficulties slider up."

Participant 2

For P2, they enjoy the challenge that comes with some achievements, a difficulty that does not require increasing the game's difficulty setting if one exists. They are motivated to prove their skill and find the challenge exciting. If it is not challenging, they will probably not go for it or will feel less excited about it. P7 and P8 share similar thoughts, where for P8 the challenge itself in achievements is more

enjoyable than the achievements themselves. P7 feels that the harder it is, the more motivated they become, and the more rewarding it feels once completed due to the effort invested.

P4 comments that even if difficulty can be fun, it depends on the type of difficulty. Adding more hours to complete something is less attractive; for example, defeating a boss 100 times is not ideal, whereas defeating a boss without armor sounds more appealing. P3 also had specific comments regarding skill and difficulty in multiplayer games, where achievements involve winning matches against real players. This is less enjoyable for them because it can involve playing against people who know the game too well or are on a different skill level.

5.8 Completion

For all participants aside from P1 and P4, the concept of completing all achievements is associated with a positive feeling. The deviation in P1 and P4 is due to them having less experience or never having completed all achievements in a game, either because they moved on to the next game before completing them, have only 100% completed a demo previously (P1), or because they only go for the achievements that interest them (P4).

As previously established, achievements are pursued in games the participants already enjoy, and they become a way of experiencing more of what the game has to offer, with the completion of achievements bringing a sense of getting closer to seeing everything the game has to offer, P2. P3 and P6 may already like a game, but once achieving 100%, they feel truly satisfied that they have invested their time into the goal and feel proud about it, as it serves as a form of proof of how much they enjoyed the game. This can also relate to money investment, feeling that they have gotten their money's worth (P2), as well as a way of being able to put the game aside and say that they have finished all there is to see (P5).

P2, P5, and P7 all commented on the Steam completion ribbon when asked about completing games, noting that they like the ribbon and associate it with "games being finished." When completed on Steam, it is added to a "perfected games" folder and a ribbon appears next to the achievement list on the game's activity page. For P8, one aspect is that it becomes a form of collecting the game through completion, which does not necessarily function as a status symbol but rather as a personal collection of accomplishments, where it is satisfying to see a fully completed checklist.

Even before completing a game, P8 likes to see the percentage of achievements completed and watch it increase toward completion, while also feeling discomfort when the percentage is close to complete but not fully complete. P6 expresses something similar, where the final remaining achievements create frustration, but also strong motivation to complete them, even leading them to invest more time than they normally would, including engaging in more grind-heavy aspects.

"I can be a completionist in a way where like sometimes you don't always enjoy being like completing a game but you still feel like you need to because you're so close so you like sometimes I have had you know final experiences with a game that might not be as fun as I would have wanted it to be because I've enjoyed a game overall but I need to get those few tedious things to complete the game."

Participant 6

If a game has achievements and P7 has started playing it, they will pursue them up to the point where they are finished with what is available in that version of the game, but they have limits regarding what they are morally okay with. If achievements are tied behind DLC, there is also a financial

investment required to complete the game fully, which raises the question of paying money for achievements rather than for additional gameplay content. This becomes acceptable when the player wants more of the game itself, but not when it is solely for achievements (P7). A similar situation occurred for P3, who completed a game fully only to discover that a DLC had been released, changing the game from perfected to uncompleted.

"It's like if I do achievements in games that I know have achievements that I can't reach without paying more money, which I want to see. Then it's a balance between wanting to 100% what I have and that I won't get that little badge when I 100% that game. Because there is a paid DLC that gives more achievements that I maybe can't reach since I don't want to like spend."

Participant 7

5.9 Relationship to the game after completion

P7 showed hesitancy towards idle games and played games like these for another purpose: a more pure dopamine-rush experience from achievements or rewards that follow the simple mechanics often found in idle games. Once completed and all achievements are received, it loses its purpose, and they instead move on to another idle game with achievements. P7 said this is due to idle games usually having similar gameplay, and the attachment they build to idle games is not as strong because of their mechanics, with the focus being more on killing time with the popping sound of achievements that the game provides. If it had more replay value, it might be replayed, but at that point it might also change the reason for why they seek out idle games to begin with.

P2, P6, and to some degree P7, as established, may still replay games after 100% completing them depending on the game and genre, but P6 admits that the effort that sometimes comes with fully completing a game results in replaying games so often that they can become somewhat tired of them for a while.

When it comes to P2, P5, and P8, they instead believe it is more likely that they will not replay the game after completion, although this again depends on the game and genre. P2 might return if the game is easily repeatable, for example if it has mechanics that are unpredictable; in those cases, there is still more for the participant to experience in the game. P5 finds themselves coming back to games to relive the story, but usually sees themselves not returning to the game after completion.

P1 and P4 are excluded from this section due to having less experience in completing games fully.

5.10 Planning the route

"Because sometimes you do like, I have relied on guides many a time in my life where you know, you check in with another person how they did something and it helps you also get that goal."

Participant 6

Some achievements require more planning and skill, either through a more planned route in the game if it is about time investment or by reading a guide. P2 groups achievements into a kind of playthrough or playstyle direction in order to set up a plan to see which ones can be done in which playthrough and when they need to be completed. Some require fresh starts, for example difficulty-based achievements, while others can be picked up along the way. P5 treats difficult achievements as main goals and easier or smaller ones as "I'll grab these on the way," and if they are

too difficult they are placed in a category of being avoided as much as possible. P5, as well as P7, also have achievements that are placed in a backlog for later, pushing the goal further ahead.

For P3, some achievements feel more meaningful because they require more steps to complete. Something that is not always planned in P3's route when going for achievements is when certain collectibles that give rewards lock some behind progression, meaning that if too much progress is made, the player cannot return to collect them without replaying the game. For them, a good achievement system is one where it is always possible to return and collect them at a later time.

5.11 Guiding

Seeing everything the game has to offer is a recurring subject when it comes to the completion mindset. What is less discussed is that this completion mindset can exist for a player without reaching 100% achievements; for some, this can extend to completing all side quests instead or unlocking all abilities. Because of this, some participants (P2, P4, P5, P6, P8) view the list of achievements both as, or instead of, a way to navigate the world, indicating where to go to see more of the game world or experience more of the story.

"For games that I end up really liking I do want to find as much of it as possible, you know, try to experience as much of it as possible. And achievements tend to be a good guide for that."

Participant 6

P1 is more relaxed regarding the concept of achievements, but even here the rewards affect the mindset. When receiving an achievement for exploring all areas, they felt a lack of interest in the game, because for them exploring was the highlight of many games. For P1 a good achievement system is viewed as something that helps guide the player toward directions and places they have yet to explore.

P5, when playing more sandbox-like games, uses achievements as goals or tasks during gameplay, finding purpose in the game through the achievements, as otherwise it is easy for them to become bored due to a lack of objectives.

5.12 Amount of achievement

Participants were asked about what the amount of achievements meant to them, this included what they roughly viewed as many and few achievements, as well as their view of it. Asking the participants what is many achievements and what is a few achievements in their view leads to variation of answers (table 3).

	P3	**P5**	**P6**	**P7**	**P8**
Few	12	5	10	25	10
Many	50	15	50	100	20

Table 3: List of what some of the participants viewed as many and few achievements in a game

P1, P2, and P4 did not give an estimate of the amount, but instead provided valuable information about how they view the number of achievements depending on the game, something the other participants also added to. The number of achievements might vary depending on factors such as game scope and genre (P2 and P4). For example, it would be too many if a four-hour game had 200 achievements, but it would be understandable if a game with 300 hours of side quest content had 200 achievements (P2). P4 sometimes feels that few achievements can make the game feel incomplete, for example, if an AAA game only had a few achievements, it would feel off, as if potential had been lost. Less achievements make the game feel smaller before properly playing it (P4).

"So, quantity has not very big impact, it is more quality, I would say."

Participant 8

Usually, the number of achievements is tied to game length, says P7. P7 typically plays both shorter and longer games, and depending on mood and available time, that determines which game is chosen. P2 and P3 have differing views on the concept of few achievements, while P3 prefers them because it makes them easier to collect, P2 can feel that there is a lack of care and thought, or alternatively that too much focus has been placed on only a few achievements or single elements of the achievement system.

In contrast to many achievements, P6 instead looks for what seems achievable. They rarely look at a list of, for example, 50 achievements and feel that it is possible to complete all of them, as there are often many different skills required the more achievements there are.

"I think that requires you to be really good at the entire, like all, everything in the game."

Participant 6

P6 comments that games with many achievements might also more commonly require players to replay the game multiple times to complete all of them, reinforcing the idea that you really have to like the game to want to do that or have the energy to pursue it multiple times. On the other hand, a large number of achievements can indicate that there are many secrets, story elements, or high-skill challenges in the game, or at least suggest that there might be, which can be exciting, P6.

P2, P5, and P7 do not appreciate it when there is a large number of achievements where players are rewarded for nearly everything in the game, as it makes them feel meaningless and diminishes the value of each achievement. This also applies if achievements start popping up constantly every other second, according to P2.

5.13 Time investment

Whether an achievement has a short time investment or a long time investment, people have preferences for both. When it comes to achievements that take a short time to complete, P8 says it does not sound as fun to complete, as it sounds like an achievement that would not involve any real challenge. P6 also adds that while an achievement with a longer time investment does not necessarily require effort, for example, reaching a certain point in the main story, when achievements are too quick or do not require much time investment, they tend not to feel as excited about them.

For P2, short time investment achievements instead function as a way to get the dopamine flowing. The same applies to P3, who says they like them and finds it a positive feeling to walk around in the game and hear them unlock. Even in this case, P2 notes that if they are too frequent or too easy, it can defeat the purpose of them being achievements. P5 points out that it might depend on whether you actually have to do anything beyond simply playing the game to get the achievement. It can still be a positive surprise to receive an achievement even if it is easy to obtain. For P5, it still feels meaningful if it involves specific interactions rather than generic progress-based triggers such as "do this, do that, reach this point."

"I think the non-meaningful achievements are the ones that you complete by doing nothing. Like oh you finish a prologue, you finish a first chapter, you finish a second chapter, like it's nice. But it is also everyone who reached this part of the game will get this one, so it doesn´t really feel like an achievement."

Participant 5

P7 and P8 agree that even short time investment achievements can feel meaningful if they require stepping out of normal gameplay. An example from P7 that is not meaningful is achievements such as finding the main menu and does not feel like real completion of something substantial.

When it comes to longer time investment achievements, P2 divides them into two categories: goal-based and grind-based. Goal-based achievements are those at the end of a playthrough or route, where other achievements can be completed along the way. Grind-based achievements are those where players must spend a large amount of time repeating missions or collecting resources, with no immediate reward structure beyond progression itself. P2 prefers goal-based achievements, as they feels they no longer has time in their life to pursue grind-heavy ones. P7 instead finds grind-based achievements enjoyable, as they are worked on gradually throughout gameplay, creating a sense that they will eventually be completed over time. They view them more as passive goals than active ones and take their time with them.

"Fun! And there are many achievements that are quite varying. They have some kinda grindy achievements, which I like."

Participant 7

P3 describes a scenario where they feel frustration when achievements involve too many random elements affecting time investment. For example, fighting all versions of a monster where spawn locations are randomized creates uncertainty about when completion will be possible. They feels it is not worth their time if an achievement is too grindy and not enjoyable, especially when the reward is minimal, such as a single achievement.

P5 tends to go for longer achievements if they feel achievable and do not require excessive time investment. This changes slightly when the achievement is tied to a game mechanic or part of the story they really enjoy. For them, the most important factor is being in the right mindset, meaning they are not easily annoyed, which makes longer achievements more enjoyable to complete.

Some achievements may simply be more complex and require more time, but compared to achievements that require repeated actions, the repeated ones are considered less enjoyable, P4 notes.

For P8, when a game requires 100 hours, 10 hours, or 50 hours for completion, it feels less meaningful. It becomes a moment where they recognise they will likely not be able to obtain the achievements and feel slight frustration, even when aware that it may be tied to unrealistically high requirements. If something feels unreasonable, they are very unlikely to pursue it, this should not be confused with cases where the activity is enjoyable, as they would still invest time if it remains fun.

5.14 The implications of having specific achievements

P1, P3, P6, and P7 all struggle with achievements that have heavy implications for the story based on player choices. When an achievement is tied to a narrative decision that places a player's favourite characters in a negative or sad situation, for example an evil run, they tend to avoid the achievement. They feel more content with not completing the achievement than with completing it.

"I have a really hard time doing that, I think it is the same thing in Baldur's gate. I´ve tried to do an evil run several times and I just can't do it because I can't. Like it feels bad doing the evil choices."

Participant 1

For P3, the priority of not hurting fictional characters outweighs the priority of 100% completing the game, and similar reasoning applies to P1, P6, and P7. They feel better not having the achievements completed than being reminded of actions and choices they made in the game. P6 also describes themself as someone who plays games partly for the characters.

P6 is also generally not interested in achievements where the goal is to perform poorly or intentionally fail, so-called "bad achievements." This is partly tied to not wanting to harm characters, but also to discomfort with failing game mechanics. They give an example of an achievement where the player must actively fail multiple times, with long intervals between opportunities to do so. The requirement of intentionally performing poorly also affects the player's reputation with non-playable characters in the game. This makes them less inclined to pursue it, both due to the time investment and the implications of "failing." They believe that if this had been presented in a more lighthearted or humorous way, their experience might have been different, but as it stands, it was not a pleasant achievement to pursue.

5.15 Chapter 1, finish chapter 1

The participants have opinions on how achievement names and descriptions are written. P3 dislikes when names and descriptions for chapter-based achievements are written in a descriptive way such as "finish chapter 1" and so on, as this makes the achievements feel boring to them. Instead, they prefer to see something more engaging, such as a quote from the game when the game is linear.

"All those achievements are basically finish episode 1, finish episode 2, 3, 4 and I´m like woaw ye."

Participant 3

P2 agrees that it helps when achievement descriptions are formulated in a way that makes them interested in discovering the context for them while playing the game, whereas more boring achievements do not affect them as much.

In the case of secret achievements and decisions about what should be hidden or shown, P5 says that if they are not hidden it makes you more aware of which ones you want to pursue. A big motivator for them is also whether achievements have funny names or humorous themes. This indicates that people do look at achievements and reflect on them. P7 likes to go back to achievement lists after completing them and see what they have been able to accomplish. P8 adds that it is less enjoyable to pursue achievements outside of Steam, because it is satisfying to have them collected in one place, creating a form of record of what games you have played and completed.

5.16 Social gameplay

There are people who enjoy playing with others and people who do not, for different reasons such as differing play styles or a lack of friends who want to play the same games. P4 finds it frustrating when achievements are tied to multiplayer gameplay and cannot be completed alone, particularly when they have fewer friends playing the same games as them.

In contrast, P2 and P5 sometimes enjoy spending time with friends who also pursue achievements, specifically in games where co-op is an option. This is not limited to games where multiplayer is required for achievements, but also includes cases where achievements are the same as in single-player modes, with co-op simply making the process quicker and more collaborative. In these cases, achievement hunting becomes a social activity.

5.17 Social comparison

P1 first noticed achievements more clearly through comparing achievements with a friend and seeing that the friend had completed more in a game than them. This is something that P2, P3, P5, P6, P7, and P8 also relate to, having compared achievements with friends in specific games at different levels of emotional intensity. P4 is the exception here, but even they might discuss with friends in person if they managed to do something complex in games, which could include achievements, although they do not actively seek out information about what achievements their friends have completed.

“It becomes a bit like a pin that one would have on the bag or jacket.”

Participant 8

P8’s motivation comes both from achievements being a fun challenge and from the possibility of showing others that they enjoy the game. This is compared to a pin on a bag or jacket: it is not necessarily something others need to notice or a major source of motivation, but rather a pleasant feeling of having it on one’s profile and being able to show that one is a fan of the game. P8 reflects, however, that looking at friends’ achievements is only something they do if it is in the same game they themselves are playing, making comparison enjoyable but not extending to general interest in friends’ overall achievement lists. This is also influenced by Steam as a platform; P8 mentions not being interested in some games and their achievements before they were released on Steam, and that games on other platforms were less appealing because achievements could not be collected or viewed in the same centralized way. P5 notes that they are particularly aware of this on Steam and how easy it is to see friends’ achievements when viewing their games in the Steam library, as they appear in the game activity section.

For P6, comparing achievements with friends is a way to better understand what they enjoy, what they have in common, and how skilled they are at games. If they have perfected a game, this becomes more meaningful, as it indicates how much time they are willing to invest in games rather than necessarily relating to similar games they play themself. Similarly, P7 compares achievements in co-op games to evaluate how their gameplay differs from their own when playing the same games.

P2 and P5 describe comparison as somewhat related to "stroking one's ego," looking at whether there is a similar level of time investment in earning achievements, particularly when effort has been invested in completing them. For P3, it is more of a fear of missing out when friends mention achievements or when seeing them in the Steam activity feed, especially when it is a game they also play but is missing achievements in, and particularly when they are rare with a low global completion percentage that friends have obtained.

5.18 Global percentage

Another very visual aspect of Steam that represents human interaction with achievements is the global percentage shown next to each achievement, indicating how common or rare it is. For P5, this does not really affect them, and they mostly only look at the list to get a sense of which achievements are easiest to obtain. For P1 and P8, this is not a major influence either, but it is still a source of enjoyment when seeing a low percentage, as they experience a positive feeling from being part of a smaller group of players who have completed it.

For the other participants, it is clear that it does have some influence, where lower percentages tend to make them feel better. P7 points out that it serves as a testament to having continued where others gave up in order to reach it.

"It feels extra good to have done something that you know that not a lot of people have."

Participant 6

P6 has a story about an achievement that was mostly luck-based but also slightly skill-based, and in that case it felt extra rewarding because of its rare circumstances, as well as the visual "shine" effect Steam uses for rare achievements. It encouraged them to learn more about why the achievement was rare.

"I want to say fame, but it is not really fame, nobody really cares about it, but the feeling of accomplishment after doing something few people have. So I guess feeling special."

Participant 4

For P4, they sometimes find themselves going down the achievement list to the rarest ones and specifically pursuing those first, as a way to prove their skill and gain motivation from it. P2 feels something similar when the Steam percentage is low enough to trigger the shine effect around the achievement, indicating rarity and reinforcing the joy of challenging oneself.

"Usually also means, depending on how far into the game I can also just assume that not many players played through this part. And just dropped it after the first achievement."

Participant 3

While P3 also feels enjoyment from rare achievements and low percentages, they are also more sceptical, initially assuming that the low percentage is due to players not progressing far enough in the game or losing interest early, with the overall engagement and time investment in the game affecting the percentage level.

5.19 Other rewards

A multitude of rewards at the same time can feel even more rewarding. P3 finds that some achievements feel more meaningful because they may lead to additional content, such as cutscenes, when followed. For P4, it becomes more important that rewards align with the game itself and not only with achievements, particularly when the player goes out of their way to obtain something or completes more advanced gameplay requirements. In those cases, they prefer additional rewards such as progression items or story elements.

P6 also describes a scenario where they once had only one achievement left that took an extended time to complete. While this was frustrating, it was also very rewarding, and the game provided an additional cutscene upon completion, offering rewards beyond the achievement itself.

Regarding progression items and similar systems, P2 enjoys when items and equipment are tied to in-game achievements, where the achievement sound reinforces a positive feedback loop. However, P2 can view achievements negatively if they carry inherent competitive value, for example if an MMORPG would increase a player's stats based on the number of achievements completed. This would create an unbalanced experience between achievement hunters and players who simply want to play the game.

5.20 Reflections on the achievement system

P2, P4, and P5 value achievements more when it feels like there has been thought put into them. P4 feels that if achievements are added as an afterthought, they do not really contribute to the experience; while they may not degrade the game, they also do not add anything to it. For P2, achievements should be a conscious choice made by the development team and come from a place of understanding of the game.

P6 describes a "good" achievement system as one where players have to do something in the game that is, in some sense, optional. They also feel that it can be somewhat discouraging when a game does not have achievements at all.

P7 sees a good system as one that includes an assortment of different types of achievements. Even in linear games, it is positive to have achievements tied to progression, such as completing chapters, combined with achievements that encourage challenge-based tasks or actions that require the player to go out of their way to play in specific ways.

5.21 Summary

Following after this is table 4 with a summarization of the thematic analysis and the most prominent result that came from the participants.

Enjoyment Comes First	● Players only actively pursue achievements in games they already enjoy. ● Achievements enhance engagement rather than create it. ● Most players check achievement lists after deciding they like a game.
Achievements Create Positive Emotions	● Achievements evoke accomplishment, excitement, happiness, and satisfaction. ● Feedback systems (pop-ups, sounds, progress indicators) amplify positive emotions. ● Difficult achievements provide the greatest sense of reward upon completion.
Missing or Unobtainable Achievements	● Frustration stems from bugged, missable, inaccessible, or excessively difficult achievements. ● Some players feel disappointed when they perform something impressive but receive no achievement.
Achievements Increase Replayability	● Achievements increase replayability by revealing content and encouraging additional playthroughs.
Achievements Change How People Play	● Achievements encourage experimentation with strategies, characters, mechanics, and narrative choices. ● Many participants viewed achievements as a way to experience more of a game.
Frustration Is Both Negative and Rewarding	● Tedious or difficult achievements can cause frustration. ● Overcoming frustration often creates the strongest sense of achievement. ● Players differ in how much frustration they are willing to tolerate.
Challenge Is a Major Motivator	● Many players enjoy achievements that test skill, exploration, or creativity. ● Challenges are often valued more than simply increasing game difficulty. ● Extremely difficult achievements can discourage participation.
Completion Is Highly Satisfying	● 100% completion is associated with pride, satisfaction, closure, and dedication. ● Completion serves as evidence that they fully experienced the game.
Achievements Help Guide Gameplay	● Players often use achievements as Goals, Checklists, Exploration tools ● Achievements help reveal content they might otherwise miss.
Quality Matters More Than Quantity	● Participants generally cared more about meaningful achievements than the total number. ● Too many easy achievements reduce perceived value. ● Achievement counts should match the game's scope and length.

Time Investment Matters	● Players prefer achievements that feel meaningful for the time invested. ● Goal-oriented long achievements are preferred over repetitive grinding. ● Extremely time-consuming achievements can discourage completion.
Story-Related Achievements Can Create Moral Conflict	● Some players avoid achievements requiring Evil choices, Character betrayal, Intentional failure ● Emotional attachment to characters often outweighs completionist goals.
Achievement Design Influences Appeal	● Players dislike boring achievements such as "Complete Chapter 1" ● Creative names, humor, mystery, and thematic descriptions increase interest.
Social Factors Matter	● Achievements facilitate competition, cooperation, and social comparison. ● Steam's visibility of friends' achievements can increase engagement.
Rare Achievements Feel More Valuable	● Low global percentage achievements are perceived as more prestigious and rewarding. ● Rarity often signals dedication, skill, or persistence.
Additional Rewards Increase Value	● Achievements feel more meaningful when paired with tangible items or new content ● Players dislike achievements that provide competitive advantages.
Characteristics of a Good Achievement System	Participants described a good achievement system as one that: ● Encourages exploration. ● Supports different playstyles. ● Offers optional challenges. ● Rewards meaningful actions. ● Includes both progression and challenge achievements. ● Is thoughtfully designed rather than added as an afterthought.

Table 4: Key findings from thematic analysis

6 Discussion

This paper has delved into a couple of different themes to answer the question, “What motivates players to continue playing a game in order to collect achievements”, leaning heavily on player motivation. Through the findings from the background, case study and participants it points towards (1.) the players get motivated through different means and some motivations being more prominent than others for example challenge, exploring and completion. (2.) The participants' player experiences with achievements, an indication of how much you have seen of the game. (3.) Social influences, in the case study it is shown that Steam has social functions such as comparing achievements and customization, this was something some of the participants valued as well or were aware of. (4.) Achievement design factors, some achievements were viewed as more meaningful towards the participants depending if it felt like effort had been put into it.

The side goal such as the purposes achievements can serve for researchers and game developers has also been briefly mentioned with examples of how data around achievements have been previously used and as well giving the developers insight into where the achievements appear on steam through the case study, as well as player perspectives on the subject.

6.1 Player types

Fritz and Stöckl (2023) examined motivating and desirable aspects for players in video games, some of which are very similar to achievements, such as challenging oneself, exploring, completing tasks, and, at times, escapism. Other player types can also be connected to achievements, for example pursuing achievements together in a social setting or having achievements that encourage creativity in a game or even griefing. However, based on the participants’ answers, there was a stronger resonance with challenging oneself, exploring, and completing tasks. Escapism is highlighted because participants described achievements as checklists, similar to chores. This way of thinking may also relate to a completionist mindset, where players want to check off everything on a list in order to complete everything. However, from the perspective of escapism, it may instead involve finding a new pace through the addition of new goals. Later in the discussion, challenge and exploration will be examined further, particularly how achievements can guide players to explore.

When it comes to the achievement types discussed by Jakobsson (2011) casual hunters, achievement hunters, and achievement completionists some of these concepts are designed around the Xbox console and how achievements function on that platform, as well as the fact that the article was written in 2011. Achievement systems differ between gaming platforms. Through the participants’ discussions, the case study findings, and Jakobsson’s core concepts, this paper will attempt to formulate achievement types for the game platform Steam in order to identify the core aspects that motivate the different types.

All of the interviewed participants liked the game before achievements became important to them, which differs from the achievement hunter archetype, where players may play outside of their preferred genre simply to gather points. Continuing with hunters who play less for the enjoyment of the game itself, the engaging aspect instead becomes exhibiting skill and dedication. The closest equivalent on Steam might be the global achievement percentage and comparisons with friends, allowing players to see how rare an achievement is. However, it may also simply reflect the player’s own desire to prove their ability and self-worth.

As described in the case study, Steam's overlay shows progress toward 100% completion, provides a completionist showcase, awards a ribbon for perfect games, and includes a library for completed games. This reinforces the idea that Steam encourages players to fully complete games, unlike Xbox's focus on Gamerscores. Therefore, Steam is a suitable platform for completionists who want to experience everything a game has to offer and demonstrate their appreciation for a game through the platform's completion-focused features, while also being reminded of their own hard work.

Casual achievement hunters are probably the least affected by switching between gaming platforms because, according to Jakobsson (2011), achievements for them primarily involve adding goals to a game, making achievements a secondary aspect of gameplay. What may be relevant to add from the participants' responses is the concept of achievements guiding the player, either by helping them discover more aspects of a game or by providing purpose in sandbox games through additional goals. This still falls within a middle ground where rewards matter, but not enough for the player to feel forced to continue solely because of the rewards.

Heeter (2011) also comments on mastery, referring to one's own drive to improve, as well as performance, which involves striving to perform better than others. On Steam, other players become points of comparison through friends lists and global achievement percentages. Whether a player is motivated primarily by global percentages or by an internal sense of mastery could therefore influence where they fit within these player-type categories.

6.2 Our memory of achievements

One participant in the results section mentioned how, at times, it feels as though they should receive a reward, but no achievement appears where they expected one. This, together with the preference for games that include achievements rather than those that do not, connects back to what Hodent (2021) discusses regarding how attention, motivation, and emotions influence memory. Games condition players to react positively to rewards. Noticing achievements when they appear, and allowing them to make the player feel rewarded, builds memories of positive interactions connected to those goals. This influence can become so strong that players notice when achievements are absent in situations where they expect them to appear.

Operant conditioning concerns increasing a behavior through reward and decreasing a behavior through punishment. Several participants mentioned achievement-related experiences where pursuing an achievement either did not feel rewarding or involved failure-based objectives. These experiences caused frustration and reduced the participants' desire to learn. Similarly, when game objectives became too difficult, participants described either experiencing or anticipating excessive frustration. From the perspective of operant conditioning, this introduces a form of self-imposed punishment that arises from the game demanding greater learning and skill development, to the point where participants no longer find enjoyment in acquiring the skills necessary to obtain the reward. For others, however, overcoming this frustration and achieving the required improvement in skill level becomes rewarding, serving as positive reinforcement and proof of their abilities.

Achievement systems are closely connected to activities that people learn to value early in life. Consider the following quote:

"Achievements in school are rewarded by sense of mastery of subject matter, progression through grade levels, honor roll lists, and praise from teachers and peers. Achievements in games are rewarded by a sense of mastery of the game, progression through levels, by points, prizes, inclusion on leader boards, and admiration by other players."

Heeter et al. (2011, p. 43)

Here, achievements are compared to school and learning. One participant made a similar observation, describing achievements as a smaller version of being praised for a job well done. If there is anything to learn from achievements when attempting to implement them either inside or outside games, it is that they praise the player for completing the goal associated with the achievement. This also appears to connect with people's motivation toward checklists and the satisfaction of checking tasks off, which two participants compared to completing chores.

A major focus of this research concerns the different reasons achievements may motivate players, while also introducing a distinction between intrinsic and extrinsic motivation, particularly the distinction between external and internal forms of extrinsic motivation (Ryan & Deci, 2000). Although players may enjoy the feelings achievements produce, the reason for pursuing them often originates externally. For example, would players still pursue a particular objective if the achievement associated with it did not exist, especially if it required them to play in a different way? Meaningful achievements could reasonably be placed closer to a "somewhat external" form of motivation, where pursuing the goal gradually becomes internally rewarding as the participant becomes focused on the objective itself, rather than solely on the external reward.

6.3 Visualization of achievement

The name and description of an achievement hold significance for some participants, helping them understand both the type of game they are playing and what the achievement will require from them. By reading the achievement descriptions, and possibly also considering the global achievement percentage, players form expectations regarding the time investment and skill level required. Interpreting these aspects generally also required some level of game knowledge, or at least became relevant once players knew they enjoyed the game and could understand the information provided by the achievements.

Participants also expressed clear likes and dislikes regarding achievement names and descriptions. For example, achievements with descriptions such as "Chapter 1, Finish Chapter 1" were considered less enjoyable, while participants preferred more thematic descriptions, such as quotes from the game. Reflecting on this, the issue may stem from the achievement information not aligning with the game world itself, causing the achievements to feel disconnected from the game's overall theme.

6.4 Serious games

Achievements are discussed in relation to serious games, often with the goal of improving learning outcomes through their implementation. What is less established in these discussions, but became evident through the participants in this study, is that players first need to enjoy or at least find interest in the game itself for achievements to generate engagement. This should not be confused with participants' opinions regarding sandbox-like games, where achievements can provide players with a sense of purpose. Even in those cases, however, the game still needs to be interesting to the player from the beginning. The same applies to idle games. Therefore, if achievements are to be implemented successfully in other contexts, this should be the first requirement: ensuring that participants find the core aspects of the activity at least somewhat interesting. The findings of this paper may also serve as a foundation for thinking about achievement design in serious game contexts.

6.5 Time investment and amount of achievements

Based on the participants' responses, there was a perception that games with more achievements require a wider range of skills to complete fully and encourage more varied ways of playing the game. This perception appears to contrast with the findings of Groening and Binnewies (2019), who concluded that a smaller number of achievements can lead players to perceive games as more difficult. However, this perception may change after players review the achievement list and learn how the achievements interact with the game world. Groening and Binnewies (2019) also argue that a large number of achievements can lead to players becoming accustomed to rewards, making achievements feel less meaningful. This idea resonated with the participants, who suggested that receiving achievements too frequently reduces their significance. Participants also emphasized that achievements should require at least some level of effort in order to feel rewarding.

6.6 Destructive side of achievements

Covering the subject of dark game design patterns, achievements can relate to situations that participants viewed as involving the loss of time, unexpected monetary costs, or social pressure. Although none of the interview questions specifically addressed dark game design patterns, participants still mentioned scenarios that bordered on these concepts. Some participants described finally feeling able to go to bed only after finishing the last achievement and the pressure they can put on one. In addition, participants often estimated the time investment required before deciding whether to pursue an achievement.

Money was less relevant in the participants' responses. The closest example involved wanting to obtain all achievements in a game, but needing to purchase downloadable content (DLC) that the player was personally less interested in playing, solely in order to maintain 100% completion in a game they had previously fully completed.

Loss aversion, as discussed by Hodent (2021) and Engelstein (2020), appears somewhat differently in relation to achievements because, once earned, achievements generally remain permanently attached to the player's profile. A central purpose of achievements across gaming platforms is the ability to remember accomplishments and display them to others. However, for completionists, there may still be a sense of loss aversion when new updates or DLC are released, since these additions can remove a player's previously completed status. Participants also described feeling disappointed when they were unable to complete certain achievements, particularly when they believed they should have been able to do so. In this sense, the inability to complete an achievement may itself be experienced as a form of loss.

The final aspect concerns social standing. One participant found it frustrating when achievements required playing with friends, since this depended on having friends who played the same game. In some cases, players may instead attempt to connect with strangers online, effectively "forcing" social interaction in order to complete an achievement. Another participant described needing to gain enough social endorsements and downloads on an external website to unlock an achievement. When these or similar forms of social achievements become obligations rather than optional activities, they may at times be interpreted as examples of dark game design.

6.7 Social scenarios

Participants often approached achievements with the mindset of completing all achievements, or at least as many as possible, as a way of expressing appreciation and affection for a game. For some, this appreciation was mainly personal, while for others it also involved displaying accomplishments to friends. In this sense, showing achievements to others is similar to decorating a jacket or bag with pins representing personal interests: it becomes a form of self-expression, similar to what Schell (2010) describes.

The participants' social interactions surrounding achievements also involved learning about the kinds of achievements their friends pursued. This could lead to positive social understanding and provide insight into friends' interests, but it could also result in negative judgments regarding "bad" or "evil" achievements. Another example can be found in the case study of Stardew Valley, where different achievements correspond to different routes through the game, with some options being viewed more favorably within the community. Although the participants in this study did not demonstrate clear signs of this behavior, it could still represent a potential social risk, particularly when friends are able to view one another's achievements.

People can also become interested in achievements through friends. One participant described becoming interested in achievements after noticing that a friend had obtained one in the same game they were playing, illustrating how social interaction can introduce players to achievement systems.

6.8 Challenge and guiding

One of the major motivating aspects of achievements was the challenge of becoming better at something, as well as viewing achievements as guides for exploration. However, for some participants, the challenge was not appealing at all, since any form of frustration reduced their enjoyment. Instead, these participants found motivation in how achievements guided them toward discovering more aspects of the game, encouraging exploration and completion. On the other hand, participants who enjoyed challenges were sometimes less interested in achievements that focused primarily on collecting items as a method of exploration or guidance, although most still found enjoyment in discovering secrets through achievements.

Both of these aspects, challenging the player by encouraging different playstyles and guiding the player toward hidden items or secrets, could involve either short- or long-term time investments. This also allowed players to evaluate whether pursuing an achievement was worth the required investment of time and effort.

6.9 Completion

The closer participants came to full completion, the more important it became for them to finish the remaining achievements. In addition, games with fewer achievements sometimes appeared more achievable, as participants perceived them as involving fewer different types of challenges, making completion feel more attainable.

When participants were asked whether their relationship with a game changed after completing it, their responses varied. For some, this depended on whether the game contained randomized gameplay that made each playthrough different, while others still returned to games occasionally in order to relive the story. For others, however, there was a greater likelihood that they would not return after completion. In general, participants' perceptions of the game changed, as they began to view it as

“complete,” which created a positive feeling. As Groening and Binnewies (2019) discuss the sense of closure that can accompany game completion, this appeared at least to alter how participants viewed the game, even if it did not necessarily prevent them from returning to it later.

6.10 Limitations of the study

Admittedly the transcribing was supposed to be handled with Klang.ai. Although Klang.ai produced transcripts that captured the overall meaning of the interviews, the tool frequently got words wrong and occasionally altered or omitted entire sentences. While many of these errors did not significantly affect the overall interpretation of the interviews, the incorrectly transcribed statements posed a potential risk to the responses. As a result, all interviews required manual verification against the original audio recordings. This reduced the usefulness of the automated transcription process and increased the time required for analysis. Consequently, Klang.ai was ultimately used primarily as a supporting tool for navigating the recordings rather than as a transcription solution.

The paper limits itself from focusing on specific genres in order to take a broader perspective, with the aim of achieving a more overall understanding of the subject and allowing for less narrow criteria when selecting participants.

Another potential area, or future work, is that the study only looks at individual experiences with achievements to gather perspectives. A quantitative study, or something similar, could potentially be of interest in order to gain a broader understanding of general player behavior as well, but this study had to be limited to only one method due to a lack of time, and to allow the qualitative interviews to be more thoroughly developed.

6.11 Ethical and societal considerations

While the topic of ethical and societal considerations was, to a degree, brought up in the section on the destructive side of achievements and how they can affect a player negatively or harmfully, there is also another layer to consider, where no research or groundwork has been established in the paper, such as the question of who might tend to be interested in achievements. One hypothesis is that, because achievements are framed around a “checklist” concept, they may attract people who enjoy that kind of structure or who are highly receptive to it, raising further considerations that have not yet been addressed. The possibility of achievements becoming more important than one’s own personal life is a cause for concern, particularly where it develops into a strongly immersive or escapist behavior.

6.12 Future work

One aspect currently lacking in achievement research is a comparison between different gaming platforms and the ways in which they implement achievement systems. Such comparisons could reveal the strengths and weaknesses of different approaches, as well as the kinds of player behaviors each system encourages. This research was limited to one platform and a selected theoretical background, but future studies could investigate these differences further. Similar comparative research might also be valuable for other aspects of gaming platforms, such as customization systems.

Global achievement percentages may also provide interesting opportunities for data collection in research contexts, particularly when examining which aspects of a game players find difficult to complete. This would, to some extent, be influenced by when achievements are introduced during gameplay progression. However, comparisons between achievement design, perceived difficulty, and

player skill level could provide valuable insights for future studies, as could examining which story paths players most frequently choose.

Future work could also involve comparing genre to genre on the topic of achievements to learn what is typical for the different kinds of gameplay.

7 Conclusion

This study identified several motivating factors related to specific achievements and discussed these in relation to player types, as well as how the gaming platform itself may influence player motivation. It also examined players' emotions and how memories connected to achievements can shape motivation, providing insight into what occurs when players become motivated by achievements. Furthermore, the study reflected on the research material concerning achievement visualization, serious games, global achievement percentages, time investment, the number of achievements, dark game design patterns, social scenarios, the challenging and guiding roles of achievements, and the completionist mindset.

The paper in itself ends up with player motivation, player experiences, social influences and achievement design factors to be able to answer the research question "What motivates players to continue playing a game in order to collect achievements". To finish, one last summary of achievement about the findings have been added to address what game developers could think about when making an achievement system. Some do and don't in table 5.

Does:	Don´t:
● Use achievements to enhance enjoyment ● Treat achievements as extra motivation, not the main motivation ● Reward meaningful actions, ex. exploration and challenge ● Encourage experiencing more of the game ● Variety in the achievements, ex. story progression, guiding, skill ● Make difficult achievements feel possible ● Use thematic names and description, make achievement feel connected to game world ● Respect players time, make progress feel meaningful	● Make achievements impossible, ex. time-limited, requiring unavailable players, dead multiplayers, near-impossible objectives ● Rely heavily on excessive grinding ● Missable without warning, ex. having achievement at locked areas and irreversible choices (and if added, make replayability easier) ● Overload of achievements, flooding the player with too many at the same time ● Putting the player into morally uncomfortable actions, ex. evil paths, intentional failing ● Achievement behind DLC, removes their old completion status if having one, additional money becoming required.

Table 5: Guideline for Game developers about achievements

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Appendix A - Interview Questions

Demographic:

- How do you describe your gender?
- What is your age?
- Do you have any educational or professional background related to games, research, psychology, design, or similar fields that can influence how you think about achievements?

General questions:

- What platform aside from steam do you usually play on?
- Can you describe your general gaming habits?
- Can you recall a recent game where you actively pursued achievements?
- What made you decide to pursue achievements in that particular game?
- Are there games where you choose not to engage with achievements? What influences that decision?

Motivation:

- What typically motivates you to go after achievements in a game?
- Are there moments when achievements change how you approach gameplay? If so, how?
- How do you decide whether an achievement is worth pursuing?

Emotion:

- How do you feel when you unlock an achievement?
- Can you describe a time when unlocking an achievement felt especially meaningful or not meaningful?
- How do you feel when you see an achievement you have not yet unlocked?
- Have you ever felt frustrated, indifferent, or particularly excited about an achievement? Can you describe that situation?

Completion:

- What does it mean to you to complete all achievements in a game?

- Does completing all achievements change your relationship with the game afterward?

Quantity, difficulty and time:

- How do you experience games that have many achievements compared to games with only a few?
- How does achievement difficulty influence your motivation?
- How do you experience achievements that require a significant time investment?
- How do you experience achievements that can be completed in a short amount of time?

Social aspect:

- How aware are you of other players' achievements?
- Does seeing global completion percentages influence you in any way?

Reflections:

- Are there situations where achievements negatively affect your experience?
- Have you ever felt conflicted about pursuing achievements? Can you describe that?
- How do you decide when to stop pursuing achievements in a game?
- In your view, what makes an achievement system work well or not work well?